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Maternal and neonatal outcomes associated with haemoglobin concentration in pregnancy: a Burden of Proof study

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This study evaluated dose–response relationships between haemoglobin concentration in pregnancy and maternal and neonatal outcomes based on comprehensive systematic reviews using the Burden of Proof meta-analytic framework. Risk estimates accounting for uncertainty and quantitative star ratings were generated to assess evidence strength. Our analysis revealed that even small deviations above or below haemoglobin levels with the lowest predicted risk—particularly on the low end (anaemia)—were associated with nonlinear increased risks of all-cause maternal and neonatal mortality, postpartum haemorrhage, maternal sepsis, preterm birth, low birth weight and large for gestational age. Across outcomes with at least moderate evidence strength (>2 stars), maternal haemoglobin levels associated with the lowest risk ranged from 109 to 135 g l^{-1}, challenging the appropriateness of fixed anaemia thresholds. Even modest population-level shifts in haemoglobin distributions could meaningfully affect the burden of adverse outcomes, supporting more refined, trimester- and outcome-specific thresholds to better identify at-risk individuals and guide clinical interventions.

Low haemoglobin levels impair tissue oxygen delivery, causing fatigue and reduced physical and cognitive performance[1,2]. During pregnancy in particular, this condition is associated with issues such as hypoxia, blood pressure abnormalities, oxidative stress, impaired placental function, fetal growth restriction and potential preterm labour[3–5]. Low levels of iron—an indispensable element in the function of haemoglobin, muscle and nerve cells—have additionally and independently been implicated in immune dysfunction, metabolic derangements and altered fetal neurodevelopment[6–10]. While haemoglobin levels physiologically decline during pregnancy, severely decreased levels are not normal and may either signal or lead to additional problems. Conversely, high haemoglobin levels in pregnancy may be a hallmark of poor adaptation to conditions such as pre-eclampsia[11–14] and can signal impaired uteroplacental perfusion (Box 1).

Quantification of haemoglobin as a risk factor for adverse maternal and neonatal health outcomes is crucial. The Global Burden

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BOX 1

Policy summary

Background

Understanding how maternal haemoglobin levels influence maternal and newborn health is critical for shaping effective policy. While anaemia in pregnancy is well studied, less attention has been paid to high haemoglobin levels or haemoglobin as a continuous risk factor. In this study, we screened over 29,000 unique titles and abstracts worldwide and used a robust Bayesian meta-regression framework to generate health risk estimates across the full range of maternal haemoglobin concentrations, accounting for uncertainty. We incorporated a quantitative rating system to assess the consistency of evidence for 13 key maternal and neonatal outcomes.

Main findings and limitations

Based on conservative risk estimates representing the smallest effect sizes closest to the null and supported by the data, we observed clear dose–response associations between maternal haemoglobin levels and multiple adverse maternal and neonatal outcomes (including all-cause stillbirth, a fetal outcome). Low haemoglobin (from the 15th percentile of the observed haemoglobin distribution up to the reference value) was associated with increased risks of postpartum haemorrhage (≥27%) and maternal sepsis and other infections (at least 30%)—both supported by moderate-quality evidence (three out of five stars)—and all-cause maternal mortality (≥15%) with limited (two stars) evidence. In newborns, low maternal haemoglobin was linked to elevated risks of ≥16% (three stars) for all-cause neonatal mortality, ≥3.8% for PTB, ≥3.0% for LBW and ≥5% for SGA, all with weak (two stars) evidence. In contrast, the data-dense high haemoglobin range (from the reference to the 85th percentile of the observed haemoglobin distribution) was associated with smaller increases in risks of PTB (≥2.1%), LBW (≥0.5%), SGA (≥0.22%) and all-cause neonatal mortality (≥0.1%) with limited (two stars) evidence. Sensitivity analyses confirmed the robustness of our main findings across different trimesters, study designs and reference haemoglobin values (whether model derived or using a uniform reference value of 120 g l^{-1}). These findings underscore the importance of routine monitoring and targeted management of maternal haemoglobin levels throughout pregnancy to support and optimize maternal and newborn outcomes.

Our findings are subject to several limitations inherent to the available data. Sparse data on high haemoglobin levels may hinder the precise estimation of their impact on maternal and neonatal outcomes. The lack of individual-level data limited our ability to account for contextual and confounding factors, while assumptions of consistent risk across geographies may overlook regional variation. Lastly, our model required assumptions (for example, linearity at distribution extremes) that may have influenced estimates where data were sparse.

Policy implications

Synthesized evidence shows elevated maternal and neonatal risks with both low and high maternal haemoglobin levels. Outcomes with two-star or higher evidence ratings showed minimum risk between 109 g l^{-1} and 135 g l^{-1}, suggesting that current clinical thresholds for anaemia and high-normal thresholds may miss risk-prone individuals. These minimum risk estimates represent outcome-specific reference points rather than prescriptive clinical targets and should be interpreted in conjunction with evidence strength and uncertainty. Importantly, risks varied by gestational age, with outcomes most strongly associated with abnormal haemoglobin levels occurring during earlier trimesters of pregnancy. Policies should support trimester-specific screening, revise clinical guidelines to reflect risks observed even within currently accepted haemoglobin ranges, and enhance antenatal care programmes to address nutritional deficiencies, infections and other contributors to abnormal haemoglobin concentrations—especially in low-resource regions. The conservative nature of this analysis offers strong reassurance that these risks are not overstated, supporting continued, context-targeted efforts to optimize haemoglobin concentrations, including global initiatives to reduce anaemia, to improve maternal and newborn health and promote blood health as an integral part of overall health.

of Disease (GBD) 2021 study estimated that nearly 2 billion people—disproportionately adult women and young children—were anaemic in 2021 (ref. 15). This estimate used the 2001 World Health Organization (WHO) anaemia definitions, which were based on severity thresholds initially chosen in 1968 only as the 5th percentile of haemoglobin distributions in sampled populations[16,17]. Unfortunately, most previous work evaluating adverse maternal and neonatal outcomes as a function of haemoglobin has failed to evaluate the dose–response relationship and has treated anaemia as a binary exposure[18–23], relying on the same 2001 WHO anaemia definitions. Select studies have described dose–response relationships with small for gestational age (SGA), low birth weight (LBW), preterm birth (PTB) and all-cause stillbirth[18,19,24–27], and some have evaluated the predictive value of the 5th percentile[28], but there is a notable absence of assessments based on increased risk of adverse health outcomes[16,28,29] and no studies to our knowledge have comprehensively evaluated a full range of outcomes or quantified the full nonlinear risk relationship across the entire domain of low and high haemoglobin levels. Of note, the WHO published new anaemia definitions in 2024, and GBD estimated that these changes alone mean that there are nearly 200 million more persons with anaemia than was previously apparent. Comprehensive dose–response assessments can help mitigate misclassification as definitions change and will facilitate more reliable population-level assessments to support clinical and policy decision-making.

Here we present a comprehensive evaluation of the risk relationships between haemoglobin concentration and maternal and neonatal outcomes (including all-cause stillbirth, a fetal outcome) using the Burden of Proof meta-analysis framework[30]. We synthesized all available eligible evidence to quantify dose–response relationships between low and high haemoglobin concentrations in pregnancy and 13 adverse maternal and neonatal health outcomes (Table 1). A comprehensive understanding of the health impacts associated with suboptimal haemoglobin levels has the potential to facilitate recognition of blood health as an independent, critical and modifiable determinant of optimal health.

Results

Briefly, we systematically reviewed global literature on haemoglobin in pregnancy and maternal and neonatal outcomes (including all-cause stillbirth, a fetal outcome) from January 1980 to July 2025, without location or language restrictions. We applied Burden of Proof meta-analytic models to generate conservative effect estimates—relative risks (RRs) closest to the null, adjusted for study-level covariates and between-study heterogeneity. These are expressed

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Table 1 | Strength of the evidence for the relationship between maternal haemoglobin, measured at any point during pregnancy, and maternal outcomes

Outcome	Number of unique studies (low haemoglobin, high haemoglobin)	Reference haemoglobin value ($g\,l^{-1}$)	Mean RR across exposure levels (95% UI with gamma)						15th–85th %ile of exposure	Low haemoglobin		High haemoglobin		Pub. bias	Selected bias covariates
			Severe anaemia (Hb<70 $g\,l^{-1}$)	Moderate anaemia (Hb 70–100 $g\,l^{-1}$)	Mild anaemia (Hb 100–110 $g\,l^{-1}$)	Hb 110–120 $g\,l^{-1}$	High Hb>140 $g\,l^{-1}$	High Hb>160 $g\,l^{-1}$		Mean BPRF (ROS)	Star rating (min. % excess risk)	Mean BPRF (ROS)	Star rating (min. % excess risk)		
All-cause maternal mortality	5 (5, 2)	109	2.54 (1.14–5.68)	1.76 (1.08–2.86)	1.05 (1.01–1.09)	1.07 (1.01–1.14)	NA#	NA#	65–131	1.15 (0.14)	2 (14.56)	NA#	NA#	No	NA
Postpartum haemorrhage	71 (71, 4)	135	6.70 (2.13–21.08)	2.88 (1.52–5.46)	1.44 (1.16–1.79)	1.10 (1.04–1.17)	1.06 (1.02–1.09)	1.11 (1.04–1.19)	79–129	1.27 (0.24)	3 (27.11)	NA*	NA*	Yes	Study design, method of outcome diagnosis, outcome definition, adjustment for demographic factors, blood sample type, mode of delivery
Antepartum haemorrhage	3 (3, 1)	126	1.21 (0.66–2.22)	1.21 (0.67–2.20)	1.16 (0.73–1.82)	1.06 (0.89–1.26)	NA#	NA#	89–135		0 (NA)		0 (NA)	Yes	NA
Pre-eclampsia	13 (12, 5)	123	1.54 (0.16–14.48)	1.45 (0.21–9.95)	1.30 (0.33–5.17)	1.13 (0.61–2.10)	1.90 (0.60–6.06)	2.33 (2.09–2.61)	94–141	0.52 (−0.66)	1 (NA)	0.54 (−0.61)	1(NA)	No	Multiple exposure levels, adjustment for demographic factors
Maternal sepsis and other infections	16 (16, 0)	132	5.36 (1.09–26.36)	3.32 (1.06–10.37)	1.84 (1.03–3.29)	1.25 (1.01–1.55)	NA#	NA#	63–133	1.18 (0.16)	3(17.59)	NA#	NA#	No	Adjustment for maternal comorbidities, blood sample type
Peripartum depression	13 (13, 0)	128	1.59 (0.79–3.22)	1.50 (0.81–2.78)	1.25 (0.89–1.75)	1.06 (0.97–1.16)	NA#	NA#	87–129	0.95 (−0.05)	1 (NA)	NA#	NA#	No	NA

Hb, haemoglobin; Ref. Hb, reference haemoglobin; RR, relative risk; UIs, uncertainty intervals; BPRF, burden of proof risk function; ROS, risk–outcome WHO; Min. % excess risk, minimum excess risk in per cent; Pub. bias, publication bias; NA, not applicable. The number of unique studies listed includes the total number of studies in the meta-regression, as well as those in which the haemoglobin levels of the exposed group were either lower (that is, low haemoglobin) or higher (that is, high haemoglobin) than those of the nonexposed/reference group. The reference haemoglobin value represents the exposure level that corresponds to the lowest risk value determined by the risk function. The RR of the outcome is averaged across exposure ranges (including conventional anaemia cutoffs), with uncertainties adjusted using a gamma method to account for between-study heterogeneity. The mean BPRF represents the weakest association consistent with available data (defined as the 5th-quantile risk curve closest to the null or an RR of 1) averaged across the data-dense exposure ranges below and above the reference haemoglobin value, separately, after accounting for between-study heterogeneity. The data-dense exposure range below the reference haemoglobin value is defined as the range from the 15th percentile of the haemoglobin distribution up to the reference haemoglobin value or between the 15th and 85th percentiles if the entire distribution falls below the reference value. The data-dense exposure range above the reference haemoglobin value is defined as the range from the reference haemoglobin value to the 85th percentile of the haemoglobin distribution. The ROS represents the signed value of the log(BPRF) averaged across the data-dense exposure ranges below and above the reference haemoglobin value, separately, directly comparable across outcomes. Minimum excess risk reflects the most conservative estimate of increased risk associated with haemoglobin levels that is consistent with the available data. For ease of interpretation, we have transformed the ROS and BPRF into a star rating (1–5), with a higher rating representing a larger effect and stronger evidence. Potential publication bias, which affects the validity of the results, was tested using Egger's regression. Observations were bias-adjusted to better reflect the gold-standard values of selected bias covariates, identified through an algorithm that systematically detects covariates associated with significant bias across included studies. 'NA#' indicates that mean RR estimates and evidence scores were not reported because they were derived from fewer than three studies. 'NA*' indicates that mean RR estimates and evidence scores were not reported because the entire data-dense exposure range (that is, the 15th–85th percentile of exposure) fell below the reference haemoglobin value. For outcomes with a zero-star rating, BPRF, ROS and minimum excess risk are not reported; for one-star ratings, only minimum excess risk is not reported.

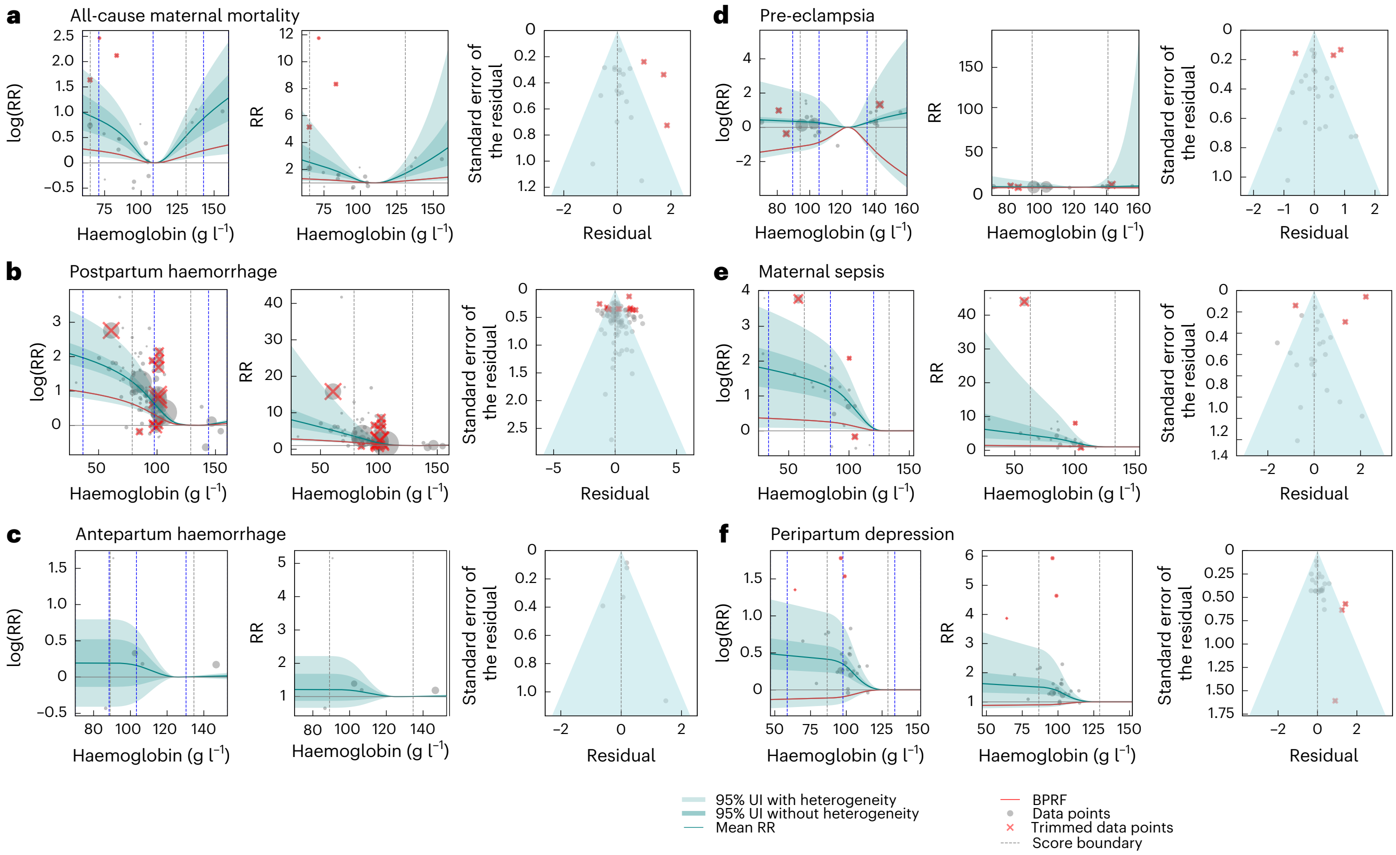

Fig. 1 | Relationship between maternal haemoglobin exposure and maternal outcomes. a–f, The relationship between maternal haemoglobin exposure (measured at any point during pregnancy) and maternal outcomes: all-cause maternal mortality (**a**), postpartum haemorrhage (**b**), antepartum haemorrhage (**c**), pre-eclampsia (**d**), maternal sepsis and other infections (**e**) and peripartum depression (**f**). For each outcome, the left and middle panels display the log-transformed RR (log(RR)) and RR functions, respectively, computed relative to an outcome-specific reference haemoglobin concentration corresponding to the level estimated to have the minimum risk. The solid dark curve shows the estimated mean RR across maternal haemoglobin levels, while the solid red curve denotes the burden of proof function. Light and dark shaded regions indicate 95% UIs excluding and including between-study heterogeneity, respectively. Grey dashed vertical lines indicate the 15th and 85th percentiles of the maternal haemoglobin concentration. Blue dashed vertical lines denote the locations of interior spline knots, which were placed based on the density of available data to flexibly capture nonlinear dose–response relationships. To visualize log(RR) points in the left panel, each data point is plotted with the *x* value at the midpoint of the alternative exposure group and the *y* value offset by the difference between the reported and predicted log risk. In the middle panel, values were exponentiated to obtain the RR curve. The right panel shows a modified funnel plot with residuals (relative to 0) on the *x* axis and the estimated standard error (including reported uncertainty and between-study heterogeneity) on the *y* axis.

as the burden of proof risk function (BPRF), representing the lowest RR consistent with the evidence, and converted into a risk–outcome score (ROS, log(BPRF)) with a one- to five-star rating reflecting the strength of the evidence and certainty. BPRFs, ROS and star ratings were estimated only within data-dense haemoglobin ranges—low (15th percentile to the reference value) and high (reference value to the 85th percentile)—where the evidence was robust. Reference haemoglobin denotes the level with the lowest risk. All 95% uncertainty intervals (UIs) incorporate both explained and unexplained heterogeneity via the gamma parameter. Publication and reporting bias were assessed using Egger's regression and visual inspection of funnel plots.

Maternal outcomes

We screened 20,192 unique titles and abstracts and 1,538 full texts, extracting 214 articles (Supplementary Figs. 2–5 and Supplementary Tables 7–10). Study designs were prospective cohort ($n = 72$), retrospective cohort ($n = 68$) and case control ($n = 47$) and studies geographically spanned 48 countries, with the largest number from India ($n = 33$), China ($n = 22$), Ethiopia ($n = 15$) and Indonesia ($n = 12$; Extended Data Fig. 1). Sufficient evidence was extracted for dose–response assessment of postpartum haemorrhage, maternal sepsis and other infections, all-cause maternal mortality, peripartum depression, pre-eclampsia and antepartum haemorrhage (Table 1 and Fig. 1). Other important maternal conditions had insufficient data (for example, peripartum cardiomyopathy, severe pre-eclampsia, eclampsia and HELLP syndrome) or lacked direct causal links to maternal haemoglobin (for example, gestational diabetes, obstructed labour; Supplementary Table 6).

Of note, only five studies across all outcomes examined associations with iron deficiency. All five studies were for peripartum depression as an outcome, but, even then, the use of different biomarkers (for example, ferritin, hepcidin, serum iron, soluble transferrin receptor) to define iron status precluded pooled analysis. No studies evaluating iron deficiency addressed other maternal outcomes.

All-cause maternal mortality had a two-star evidence rating with low haemoglobin ($n = 5$ studies) and insufficient data for high haemoglobin ($n = 2$). The lowest risk level was at 109 g l^{-1} with a mean RR of 1.07 (95% UI, 1.01–1.14) at 'low normal' haemoglobin levels (110–120 g l^{-1}), rising to 2.54 (95% UI, 1.14–5.68) at the WHO threshold for severe anaemia (Table 1, Fig. 1a and Supplementary Table 25). No bias covariates were retained (Supplementary Table 41). No publication bias was detected.

Postpartum haemorrhage had a three-star evidence rating with low haemoglobin ($n = 71$ studies), with minimum risk at 135 g l^{-1} and mean RRs of 1.10 (1.04–1.17) for 'low normal' haemoglobin and

6.70 (2.13–21.08) for severe anaemia (Table 1, Fig. 1b and Supplementary Table 25). Metrics were not generated for high haemoglobin because the data-dense exposure range fell entirely below the reference level, but the mean RRs at >140 g l^{-1} and >160 g l^{-1} were 1.06 (1.02–1.09) and 1.11 (1.04–1.19), respectively. Covariates capturing study design, outcome diagnosis method, outcome definition, blood sample type, mode of delivery and adjustment for demographic factors were retained in the final model (Supplementary Table 42). Publication bias was detected. Antepartum haemorrhage evidence was sparse, with no significant relationship for low haemoglobin (n = 3 studies) and insufficient data for high haemoglobin (n = 1; Table 1) (Fig. 1c and Supplementary Table 25). No covariates were selected in the model, and no publication bias was detected (Supplementary Table 43).

Hypertensive disorders of pregnancy (HDOP) were analysed as two separate conditions: gestational hypertension and pre-eclampsia. Traditional meta-regressions indicated no significant association between low haemoglobin and gestational hypertension (Extended Data Fig. 2). Pre-eclampsia was found to have a weak one-star evidence rating for both low haemoglobin (n = 12 studies) and high haemoglobin (n = 5), with a reference level of 123 g l^{-1} and high heterogeneity across studies (Table 1, Fig. 1d and Supplementary Table 25). Covariates for exposure level and inadequate control of demographic confounders were retained (Supplementary Table 44). No publication bias was detected.

Maternal sepsis and other maternal infections had a three-star evidence rating for low haemoglobin (n = 16 studies), no data for high haemoglobin (n = 0), estimated minimum risk at 132 g l^{-1}, and mean RRs of 1.25 (1.01–1.55) and 5.36 (1.09–26.36) for 'low normal' haemoglobin and severe anaemia, respectively (Table 1, Fig. 1d and Supplementary Table 25). Covariates capturing adjustment for maternal comorbidities and blood sample type were selected (Supplementary Table 45) with no detection of publication bias.

Peripartum depression was found to have a weak one-star relationship with low haemoglobin (n = 13 studies), with minimum risk at 128 g l^{-1} and high between-study heterogeneity suggesting potentially strong but inconsistent evidence of a relationship (Table 1, Fig. 1e and Supplementary Table 25). No bias covariates were selected (Supplementary Table 46), and no publication bias was detected. No studies reported on high haemoglobin.

Neonatal outcomes

We screened 16,100 unique titles and abstracts and 4,156 full texts, extracting 473 articles (Supplementary Fig. 6 and Supplementary Table 11). Study designs included prospective cohort (n = 157), retrospective cohort (n = 133) and case control (n = 73), and studies geographically spanned 88 countries, with studies in India (n = 79), China (n = 35), Ethiopia (n = 34) and Iran (n = 24) contributing the most observations (Extended Data Fig. 3). Sufficient evidence was extracted for dose–response assessment of all-cause neonatal mortality, PTB (<37 weeks), LBW (<2,500 g), SGA (weight in <10th percentile for gestational age), all-cause stillbirth (≥20 weeks of gestation) and neonatal sepsis (Table 2 and Fig. 2). We also considered, but excluded, haemolytic disease and neonatal encephalopathy due to lack of biological plausibility (Supplementary Table 6). Data specific to iron deficiency were again sparse, with only LBW (n = 7 studies), SGA (n = 5) and PTB (n = 12) having any data. Similarly, heterogeneity in iron status biomarkers precluded pooled analyses.

All-cause neonatal mortality had a three-star evidence rating for low haemoglobin (n = 46) and a two-star rating for high haemoglobin (n = 5). Minimum risk was at 124 g l^{-1}; the mean RR was 2.73 (95% UI, 1.72–4.33) for severe anaemia and 1.04 (1.02–1.05) for haemoglobin >160 g l^{-1} (Table 2, Fig. 2a and Supplementary Table 25). The final model included covariates capturing demographic and obstetric adjustments (Supplementary Table 47) with publication bias detected.

All-cause stillbirth had a weak one-star evidence rating for both low haemoglobin (n = 6 studies) and high haemoglobin (n = 3). The minimum risk was much lower than that for other outcomes at 95 g l^{-1} but with high between-study heterogeneity (Table 2, Fig. 2b and Supplementary Table 25). Outcome diagnosis method, representativeness and maternal comorbidity adjustment were selected by the model as bias covariates (Supplementary Table 48). Possible publication bias was detected.

PTB, LBW and SGA all showed J-shaped dose–response associations with a two-star evidence rating for both low haemoglobin (n = 146, n = 207, and n = 61 studies, respectively) and high haemoglobin (n = 30, n = 29, and n = 19, respectively). The lowest risk levels were 124, 124 and 130 g l^{-1}, and the mean RR at the severe anaemia threshold was elevated for each at 2.66 (1.16–6.11), 2.83 (1.00–7.99) and 1.62 (1.14–2.30). For haemoglobin >160 g l^{-1}, the respective mean RRs were 1.20 (1.03–1.40), 1.56 (1.00–2.43) and 1.31 (1.08–1.59; Table 2, Fig. 2c–e and Supplementary Table 25). The PTB model selected covariates including study design, blood sample type, exposure level, representativeness, selection bias, inadequate adjustment for health behaviours, obstetric factors, maternal comorbidities and socioeconomic factors; LBW bias covariates included exposure level, selection bias, adjustment for health behaviours, maternal comorbidities and demographic factors; and the SGA model selected only adjustment for socioeconomic factors (Supplementary Tables 49–51). The PTB and LBW models, but not the one for SGA, detected potential publication bias. Large for gestational age (LGA), in contrast, was data sparse with a one-star evidence rating for low haemoglobin (n = 8), insufficient data (n = 1) for high haemoglobin and the lowest risk at 131 g l^{-1} (Table 2, Fig. 2f and Supplementary Table 25). No covariates were selected in the model (Supplementary Table 52), and publication bias was not detected.

Neonatal sepsis yielded a zero-star evidence rating for low haemoglobin (n = 3 studies) and no data for high haemoglobin (n = 0; Table 2, Fig. 2g and Supplementary Table 25). The model selected no covariates (Supplementary Table 53) and found no publication bias.

Sensitivity and additional analyses

Categorical versus continuous haemoglobin modelling. For most outcomes, continuous (dose–response) haemoglobin models revealed significant associations undetected by categorical (anaemia or high haemoglobin) models, indicating that binary thresholds may obscure variation in risk (Extended Data Figs. 4–7).

Timing of haemoglobin measurement. Most haemoglobin measurements occurred in the third trimester. Trimester-specific analyses generally aligned with the overall findings (Supplementary Tables 26–31 and Supplementary Figs. 7–12), with the exception of the first-trimester associations being stronger for pre-eclampsia (two versus one star) and weaker for all-cause neonatal mortality (one versus three stars). Second-trimester exceptions were stronger evidence for PTB (three versus two stars) and weaker for pre-eclampsia (zero versus one star) and LBW (two versus one star). Third-trimester analysis showed weakened evidence for maternal sepsis (two versus three stars).

Study design and outcome definitions. Sensitivity analyses of 38 studies showed that socioeconomic adjustments did not affect the star rating for LBW (Supplementary Table 32 and Supplementary Figs. 13–17). Including studies with unclear risk–outcome temporality strengthened the evidence for pre-eclampsia (one versus three stars) but not antepartum haemorrhage.

Allowing unclear definitions did not affect PTB and SGA but strengthened the evidence for stillbirth and LBW (one versus two stars). Compared with broader outcomes, evidence for moderate PTB (<32 weeks of gestation) was similar and was strengthened (one versus two stars) for very LBW (<1,500 g), suggesting that dose responsiveness may also apply for outcome assessment. For stillbirth, additional sensitivity analyses of more restrictive gestational age thresholds (≥22, ≥24 and ≥28 weeks) yielded directionally consistent associations, with

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Table 2 | Strength of the evidence for the relationship between maternal haemoglobin, measured at any point during pregnancy, and neonatal outcomes

Outcome	Number of unique studies (low haemoglobin, high haemoglobin)	Reference haemoglobin value (g l^{-1})	Mean RR across exposure levels (95% UI with gamma)						15th–85th %ile of exposure	Low haemoglobin		High haemoglobin		Pub. bias	Selected bias covariates
			Severe anaemia (Hb<70 g l^{-1})	Moderate anaemia (Hb 70–100 g l^{-1})	Mild anaemia (Hb 100–110 g l^{-1})	Hb 110–120 g l^{-1}	High Hb>140 g l^{-1}	High Hb>160 g l^{-1}		Mean BPRF (ROS)	Star rating (min. % excess risk)	Mean BPRF (ROS)	Star rating (min. % excess risk)		
All-cause neonatal mortality	46 (46, 5)	124	2.73 (1.72–4.33)	1.47 (1.23–1.76)	1.09 (1.05–1.13)	1.02 (1.01–1.03)	1.03 (1.01–1.04)	1.04 (1.02–1.05)	69–131	1.16 (0.15)	3(16.3)	1 (0)	2(0.13)	Yes	Adjustment for demographic factors and adjustment for obstetric factors
All-cause stillbirth	6 (6, 3)	95	2.20 (0.18–27.29)	1.23 (0.63–2.43)	1.15 (0.74–1.79)	1.33 (0.53–3.32)	1.35 (0.51–3.58)	1.33 (0.54–3.29)	78–136	0.8 (−0.22)	1(NA)	0.67 (−0.4)	1(NA)	Yes	Method of outcome diagnosis, representativeness and adjustment for maternal comorbidities
PTB	148 (146, 30)	124	2.66 (1.16–6.11)	1.52 (1.06–2.17)	1.09 (1.01–1.17)	1.05 (1.01–1.10)	1.16 (1.02–1.33)	1.20 (1.03–1.40)	85–137	1.04 (0.04)	2 (3.83)	1.02 (0.02)	2(2.13)	Yes	Blood sample type, adjustment for maternal comorbidities, multiple exposure levels, adjustment for obstetric factors, adjustment for socioeconomic factors, selection bias, study design, adjustment for health behaviours and representativeness
LBW	212 (207, 29)	124	2.83 (1.00–7.99)	1.57 (1.00–2.46)	1.14 (1.00–1.29)	1.04 (1.00–1.08)	1.32 (1.00–1.74)	1.56 (1.00–2.43)	83–136	1.03 (0.03)	2(2.95)	1 (0)	2(0.46)	Yes	Study design, multiple exposure levels, adjustment for health behaviours, adjustment for socioeconomic factors, adjustment for demographic factors, timing of outcome and representativeness
SGA	61 (60, 19)	130	1.62 (1.14–2.30)	1.21 (1.05–1.38)	1.05 (1.01–1.08)	1.03 (1.01–1.05)	1.15 (1.04–1.26)	1.31 (1.08–1.59)	82–137	1.03 (0.03)	2(2.6)	1 (0)	2(0.22)	No	Adjustment for socioeconomic factors
LGA	8 (8, 1)	131	1.49 (0.76–2.91)	1.46 (0.77–2.78)	1.35 (0.81–2.26)	1.16 (0.90–1.50)	NA#	NA#	89–142	0.92 (−0.09)	1(NA)	NA#	NA#	No	NA
Neonatal sepsis	3 (3, 0)	129	3.43 (0.34–34.17)	2.09 (0.53–8.23)	1.29 (0.80–2.10)	1.08 (0.94–1.25)	NA#	NA#	51–125		0(NA)	NA#	NA#	No	NA

The number of unique studies listed includes the total number of studies in the meta-regression, as well as those in which the haemoglobin levels of the exposed group were either lower (that is, low haemoglobin) or higher (that is, high haemoglobin) than those of the nonexposed/reference group. The reference haemoglobin value represents the exposure level that corresponds to the lowest risk value determined by the risk function. The RR of the outcome is averaged across exposure ranges (including conventional anaemia cutoffs), with uncertainties adjusted using a gamma method to account for between-study heterogeneity. The mean BPRF represents the weakest association consistent with available data (defined as the 5th-quantile risk curve closest to the null or an RR of 1) averaged across the data-dense exposure ranges below and above the reference haemoglobin value, separately, after accounting for between-study heterogeneity. The data-dense exposure range below the reference haemoglobin value is defined as the range from the 15th percentile of the haemoglobin distribution up to the reference haemoglobin value or between the 15th and 85th percentiles if the entire distribution falls below the reference value. The data-dense exposure range above the reference haemoglobin value is defined as the range from the reference haemoglobin value to the 85th percentile of the haemoglobin distribution. The ROS represents the signed value of log(BPRF) averaged across the data-dense exposure ranges below and above the reference haemoglobin value, separately, directly comparable across outcomes. Minimum excess risk reflects the most conservative estimate of increased risk associated with haemoglobin levels that is consistent with the available data. For ease of interpretation, we have transformed the ROS and BPRF into a star rating (1–5), with a higher rating representing a larger effect and stronger evidence. Potential publication bias, which affects the validity of the results, was tested using Egger's regression. Observations were bias adjusted to better reflect the gold-standard values of selected bias covariates, identified through an algorithm that systematically detects covariates associated with significant bias across included studies. 'NA#' indicates that mean RR estimates and evidence scores are not reported because they were derived from fewer than three studies. 'NA*' indicates that mean RR estimates and evidence scores are not reported because the entire data-dense exposure range (that is, the 15th–85th percentile of exposure) fell below the reference haemoglobin value. For outcomes with a zero-star rating, BPRF, ROS and minimum excess risk are not reported; for one-star ratings, only minimum excess risk is not reported.

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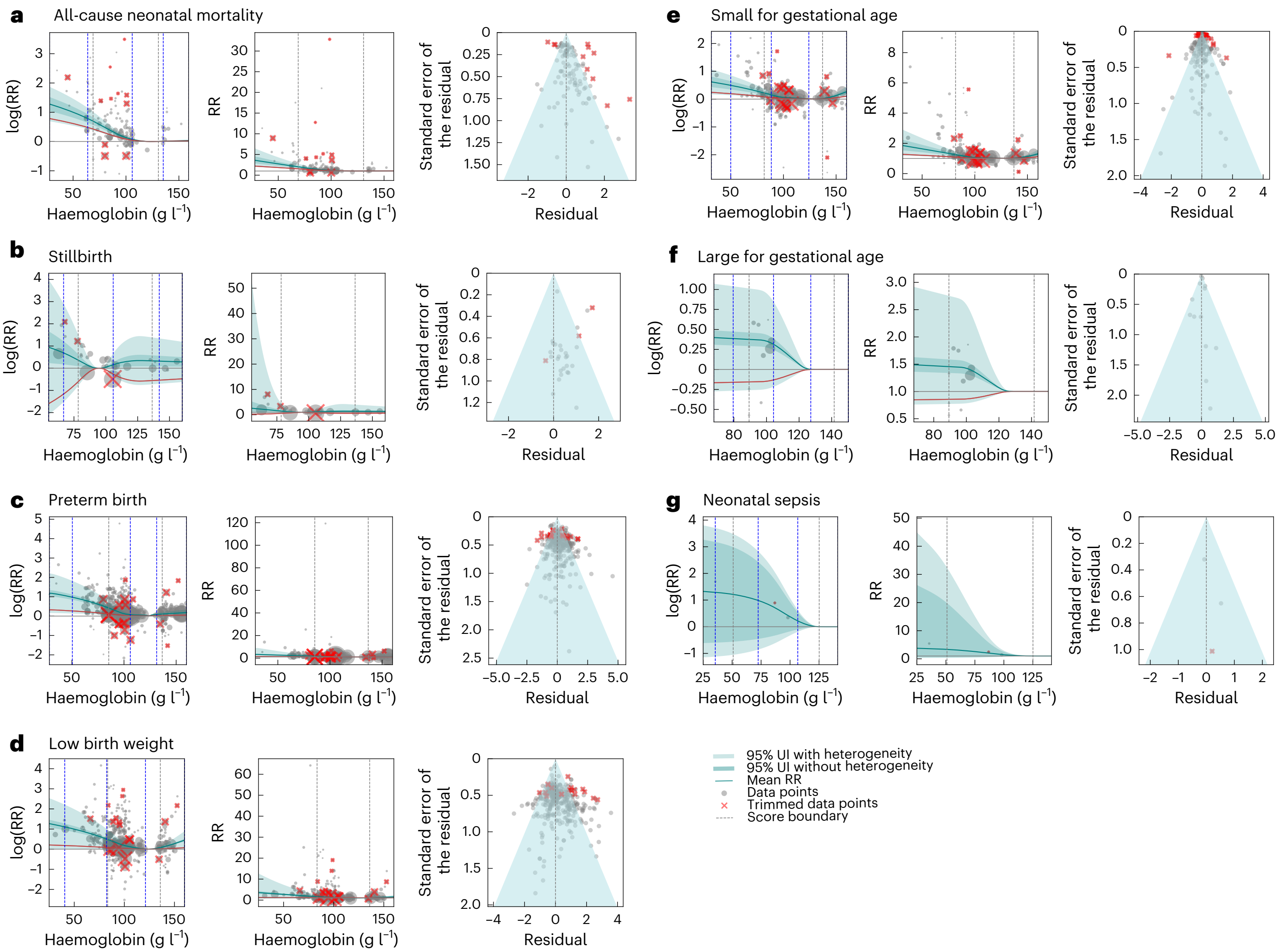

Fig. 2 | Relationship between maternal haemoglobin exposure and neonatal outcomes. a–g, The relationship between maternal haemoglobin exposure (measured at any point during pregnancy) and neonatal outcomes: all-cause neonatal mortality (**a**), all-cause stillbirth (**b**), PTB (**c**), LBW (**d**), SGA (**e**), LGA (**f**) and neonatal sepsis (**g**). Note: For each outcome, the left and middle panels display the log-RR (log(RR)) and RR functions, respectively, computed relative to an outcome-specific reference haemoglobin concentration corresponding to the level estimated to have the minimum risk. The solid dark curve shows the estimated mean RR across maternal haemoglobin levels, while the solid red curve denotes the burden of proof function. Light and dark shaded regions indicate 95% UIs excluding and including between-study heterogeneity, respectively. Grey dashed vertical lines indicate the 15th and 85th percentiles of the maternal haemoglobin concentration. Blue dashed vertical lines denote the locations of interior spline knots, which were placed based on the density of available data to flexibly capture nonlinear dose–response relationships. To visualize log(RR) points in the left panel, each data point is plotted with the *x* value at the midpoint of the alternative exposure group and the *y* value offset by the difference between the reported and predicted log risk. In the middle panel, values were exponentiated to obtain the RR curve. The right panel shows a modified funnel plot with residuals (relative to 0) on the *x* axis and the estimated standard error (including reported uncertainty and between-study heterogeneity) on the *y* axis.

evidence strength ranging from one to three stars and higher minimum risk levels (from 95 g l^{-1} in the primary analysis to 126–135 g l^{-1} for more restrictive stillbirth definitions).

Model-derived versus functional reference haemoglobin. Model-derived reference haemoglobin concentrations—defined as the theoretical minimum risk points—and the functional reference concentrations, where the risk curve begins to plateau, were consistent and aligned with normative maternal physiology (Supplementary Figs. 18–25). The only exception was for the most inclusive definition of stillbirth (≥20 weeks of gestation) where the model-derived reference was physiologically low (95 g l^{-1}; see above for stillbirth sensitivity analysis results). To evaluate the impact of using a uniform reference haemoglobin concentration, we tested a single threshold of 120 g l^{-1}—aligned with WHO's nonpregnant anaemia threshold and the median of the haemoglobin distribution at 17–18 weeks' gestation[16,31]. Results were largely consistent with the main analyses using model-derived, outcome-specific reference values (Supplementary Tables 34 and 35), supporting the robustness of findings to reference choice. An exception was all-cause maternal mortality: using the reference of 120 g l^{-1} shifted the evidence rating from three to two stars and indicated a protective association with mild anaemia, consistent with the J-shaped risk curve and the model-derived lowest risks at 109 g l^{-1}.

Discussion

We comprehensively evaluated low and high haemoglobin as dose-dependent risk factors for pregnancy outcomes using rigorous methods, finding evidence of associations with 13 different maternal and neonatal outcomes (including stillbirth, a fetal outcome). For low haemoglobin, we found limited-to-moderate strength of evidence (two to three stars out of five) that even modest reductions in haemoglobin—above current WHO anaemia thresholds—were associated with excess risk for all-cause maternal mortality, all-cause neonatal mortality, incident postpartum haemorrhage, maternal sepsis and other infections, PTB, LBW, and SGA. We found very weak (one star) evidence for pre-eclampsia, peripartum depression, stillbirth (≥20 weeks) and LGA and insufficient (zero star) evidence for antepartum haemorrhage and neonatal sepsis, primarily due to sparse or heterogeneous input data, suggesting the need for targeted investigations. For high haemoglobin, we found limited (two stars) evidence of increased risk of all-cause neonatal mortality, PTB, LBW and SGA, with very weak (one star) evidence for stillbirth and pre-eclampsia. All other outcomes did not have sufficient data to generate estimates. Sensitivity analyses largely confirmed the robustness of these findings across study designs.

The findings in this study support a concept of blood health where clinical anaemia assessments transition away from a categorical and normative view that emphasizes only severe disease and instead consider the specific health consequences of low haemoglobin, individual trends and the timing of measurement. First, adverse health impacts are evident even when haemoglobin levels remain above mild anaemia thresholds[16]. Theoretical minimum risk exposure levels (TMRELs) were mostly in the range of 120–130 g l^{-1} for outcomes with moderate or strong evidence. Second, even mild decreases from optimal levels and small changes in haemoglobin, much smaller than the changes of 10–30 g l^{-1} required to reclassify anaemia severity, are associated with meaningfully increased risk. The mean RR of maternal sepsis and other infections, for example, increases by almost 25% when the haemoglobin concentration drops from 125 g l^{-1} to 115 g l^{-1}. Such differences, especially at the population level, translate into considerable additional disease burden. Third, trimester-specific sensitivity analyses suggest opportunities for early identification of at-risk pregnancy via early and repeated haemoglobin testing. In the first and second trimester, both low and high haemoglobin have three-star associations with PTB, LBW and SGA, and low haemoglobin in the third trimester (prelabour) has three-star associations with both postpartum haemorrhage and all-cause neonatal mortality and one-star associations with maternal sepsis and peripartum depression.

All two- and three-star dose–response relationships warrant immediate consideration for clinical and public health intervention. The relationship with elevated risk of incident postpartum haemorrhage is particularly notable, as anaemic women, by definition, will have a greater risk of life-threatening consequences from haemorrhage. Our findings qualitatively align with previous evidence[18,19,27,32,33] while offering a more complete systematic review along with more robust interpretation of evidence strength through the Burden of Proof framework, which avoids categorical anaemia definitions, evaluates the full haemoglobin distribution, allows the data to determine the shape of the relationship, quantifies uncertainty related to study design and heterogeneity, and synthesizes risk into interpretable star ratings reflecting evidence certainty. This study applied location-specific WHO 2024 haemoglobin elevation adjustments[16,34], harmonizing measurements across studies and enabling direct comparability, an approach we advocate to be a standard going forward.

One-star relationships, which may change with new evidence, and zero-star relationships, which can reflect absent or sparse data, highlight priority areas for further research. Few outcomes had sufficient data to evaluate relationships with high haemoglobin, and only four neonatal and one maternal outcome had first-trimester haemoglobin measurements. Neonatal sepsis and peripartum depression exemplify outcomes with strong biological plausibility of association between low haemoglobin and increased outcome rates but suffer from sparse data, and additional studies have the potential to influence evidence assessments.

The consistency and specificity of outcome definitions are also of paramount importance. This study analysed evidence for the most inclusive stillbirth definition (≥20 weeks of gestation), finding only a one-star evidence rating and a TMREL of 95 g l^{-1}. More than half of all studies did not specify how stillbirths were defined, and a final sensitivity analysis that did not control for outcome definition, unsurprisingly, illustrated very high heterogeneity, with a one-star evidence rating and TMREL of ~101 g l^{-1}. Sensitivity analyses testing more restrictive stillbirth thresholds (≥22, ≥24 and ≥28 weeks of gestation) yielded results directionally consistent with ours, with higher reference haemoglobin levels (126–135 g l^{-1}) but input datasets similarly sparse to those in our main stillbirth assessment. New and updated assessments of the association between haemoglobin levels and stillbirth, with comprehensive reporting by gestational age threshold, have great potential not only to influence the assessment of increased stillbirth risk but also to potentially shed light on how different causes or timing of stillbirths are related to upstream risk factors.

The role of iron is another area where more data are needed. While iron and haemoglobin are closely related, they are neither synonymous nor fully interchangeable. Inadequate iron supply (that is, iron deficiency, from insufficient intake, metabolic dysregulation or excess losses) can lead to lower haemoglobin, while red blood cell destruction can cause iron overload. Beyond red blood cells, iron is also essential for muscle, immune, and nerve cells through distinct pathways[6–9]. However, most studies evaluating the health impacts of iron either reported only changes in haemoglobin or failed to adjust for haemoglobin when assessing the impacts of iron fortification, supplementation or repletion[35,36]. Although our systematic reviews concurrently sought data on the health effects of suboptimal iron independent of haemoglobin levels, we found insufficient data to inform Burden of Proof analyses for any of the evaluated outcomes owing to sparse data, heterogeneous definitions of iron deficiency and variable use of different proxy biomarkers.

While iron deficiency remains the most common cause of anaemia globally, underpinning WHO's recommendation for daily iron–folic acid supplementation[37], focusing solely on iron is insufficient, as iron deficiency accounts for only 37% of anaemia among nonpregnant women[38]. Anaemia has diverse aetiologies[39,40], including inherited blood disorders (for example, sickle cell disease, thalassaemia), chronic blood loss from abnormal uterine bleeding, menstruation or gastrointestinal losses, parasitic infections (for example, hookworm, malaria) and anaemia associated with chronic diseases (for example, cancer, kidney disease)[41,42]. Notably, increased risks observed at haemoglobin levels still within conventional non-anaemic ranges may in part reflect underlying iron deficiency, which can precede overt anaemia and contribute to morbidity[43]. The USAID Advancing Nutrition Anemia Task Force recommends adopting an ecological framework that recognizes anaemia's heterogeneous causes as well as their overlapping nature and complex interactions with external factors such as environmental, structural and social determinants[44,45].

Effective anaemia interventions at individual and population levels will require diagnostic capacity, followed by a combination of infection prevention and treatment (for example, malaria chemoprophylaxis, antihelminth treatment), chronic disease management, blood transfusions and even erythropoiesis stimulation[30,46]. Population-specific epidemiological data, such as cause-specific anaemia burden estimates produced by the GBD collaboration, can guide diagnostic and treatment priorities[47]. Continued investment in surveillance and synthesis efforts—at the global, national or subnational level—is crucial to provide timely local data on cause-specific anaemia prevalence, track progress, plan responses and capture external influences. Given the reduction in haemoglobin monitoring efforts, including large-scale programmes like the Demographic and Health Survey, ongoing population-level data collection, through the GBD collaboration, is especially critical to systematically inform anaemia burden models and improve cause-specific anaemia estimates. In parallel, coordinated policy and sustained political commitment are essential to establish and maintain such systems, ensuring blood health integration into broader maternal and child health strategies.

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Several limitations should be acknowledged. First, data constraints prevented location-specific modelling; thus, geographic, temporal or age-related risk differences cannot be ruled out. Second, the lack of data on iron status or other anaemia aetiologies limited insight into cause-specific associations and potential confounding. Third, defining anaemia using haemoglobin cutoffs necessitated imputing bounds from GBD 2021 anaemia distributions, capping haemoglobin at 141 g l^{-1} (97th percentile of haemoglobin in healthy pregnant women from the INTERGROWTH-21st population-based study[31]). This truncation may bias the threshold for high haemoglobin towards the null. Fourth, while adjusting for blood sample type, we could not account for measurement methods (for example, HemoCue), posing potential residual variability. Fifth, sparse observations at haemoglobin extremes necessitated linearity assumptions, which may underestimate risks at the tails. Sixth, limited by input data granularity and varied adjustment methods, we prioritized the most comprehensively adjusted estimates and used the Burden of Proof framework to account for confounding. We did not systematically address overadjustment, as inconsistent reporting made post hoc reclassification without individual-level data untenable. Instead, we relied on conservative evidence grading and uncertainty propagation within the Burden of Proof framework, which may affect stillbirth results, in particular, where adjustment for complications or care levels may attenuate associations. Seventh, outcome definition variability (for example, different SGA growth charts or depression scales) contributed to heterogeneity, although we applied bias covariates to mitigate this. Eighth, while we tested for publication bias, we did not adjust for it, and, relatedly, our adjustment for study clustering was limited to inflating standard errors proportional to the number of repeated measurements. Finally, our main analysis included haemoglobin measurements regardless of trimester, which may have introduced variability, although trimester-specific analyses confirmed the overall robustness of the results.

This systematic review and meta-analysis provides a comprehensive and methodologically rigorous assessment of the quality of evidence and strength of association between maternal haemoglobin levels and a wide range of maternal and neonatal outcomes (including stillbirth). Our findings challenge the current reliance on fixed anaemia thresholds and call for a paradigm shift. Haemoglobin should be recognized as a dynamic, continuous biomarker of maternal and neonatal health. Clinical guidance and public health strategies should evolve to reflect a continuum of risk, moving towards risk-based, outcome-specific guidance and prioritizing blood health, recognizing haemoglobin as one measurable component. Integrating blood health into routine public health and clinical care and strengthening surveillance systems, especially during pregnancy—a time of heightened vulnerability and opportunity for intervention—will be critical to identify, manage and ultimately prevent avoidable maternal and neonatal morbidity and mortality and advance health equity across populations.

Methods

Overview

In this study, we estimated the conservative risk of maternal and neonatal outcomes (including all-cause stillbirth, a fetal outcome) associated with low and high maternal haemoglobin levels and assessed the strength of evidence supporting these associations using the Burden of Proof methodology[30]. This methodology uses a meta-regression tool, meta-regression–Bayesian, regularized, trimmed (MR-RBT)[48], to estimate RRs and UIs that incorporate within- and between-study heterogeneity and account for systematic bias. Between-study heterogeneity that can be explained by study-level covariates (for example, differences in outcome definitions or population representativeness) is accounted for in the model, while unexplained heterogeneity is captured by the parameter gamma (γ), which inflates UIs to reflect variation not explained by measured covariates and contributes to the overall evidence strength rating.

The Burden of Proof method comprises six main analytical steps: (1) systematically reviewing and extracting data from studies reporting on the associations of interest; (2) estimating the shape of the dose–response associations between exposure and outcomes; (3) testing and adjusting for systematic sources of bias within input sources; (4) quantifying between-study heterogeneity and accounting for within-study correlation to estimate UIs; (5) evaluating publication and reporting bias using Egger's regression; and (6) estimating the BPRF, defined as the 5th-quantile estimate of the most conservative RR estimate (that is, closest to the null) reflecting the smallest harmful effect of risk exposure based on available evidence, and the corresponding ROS, defined as the signed value of log(BPRF) reflecting the strength of evidence for each association.

The Burden of Proof methodology has been detailed elsewhere[30], with the specific approaches used to estimate the association between maternal haemoglobin and maternal and neonatal outcomes described below. This study complied with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines[49] (Supplementary Tables 36 and 37) and Guidelines for Accurate and Transparent Health Estimates Reporting recommendations[50] (Supplementary Table 38).

Systematic review and data extraction

We conducted separate systematic reviews to identify peer-reviewed studies that measured the association between low or high maternal haemoglobin (that is, the exposure or risk), defined as haemoglobin concentrations below or above a certain threshold measured during pregnancy, and each of the six maternal outcomes (all-cause maternal mortality, postpartum haemorrhage, antepartum haemorrhage, HDOP (including gestational hypertension and pre-eclampsia), maternal sepsis and other infections, peripartum depression) and seven neonatal outcomes (all-cause stillbirth (fetal), all-cause neonatal mortality, PTB, LBW, SGA, LGA and neonatal sepsis). Outcomes were selected based on known epidemiological evidence of their association with low and high maternal haemoglobin[23], with definitions prespecified according to the International Classification of Diseases and other international standards (Table 3 and Extended Data Table 1). All-cause stillbirth outcomes were identified through broad neonatal and perinatal mortality searches rather than stillbirth-specific search terms; stillbirth data were extracted when reported within eligible studies. Perinatal death was accepted as an alternative definition for all-cause neonatal mortality when the timing indicated postnatal death; fetal deaths occurring at delivery were classified as all-cause stillbirth only when explicitly reported as such and meeting the ≥20-week gestational age criterion. As studies have frequently evaluated multiple neonatal outcomes, one systematic review was conducted to identify literature reporting measures of association of maternal haemoglobin and all neonatal outcomes. For the same reason, one systematic review was conducted for all-cause maternal mortality and maternal sepsis and other infections and one for antepartum and postpartum haemorrhage. Details of the peripartum depression systematic review and additional analyses have been reported elsewhere[51]. Other maternal and neonatal outcomes that were evaluated but not analysed due to inadequate data or lack of biological plausibility can be found in Supplementary Table 6.

Eight databases (PubMed, Embase, Web of Science, Cumulative Index to Nursing and Allied Health Literature (CINAHL), China National Knowledge Infrastructure (CNKI), Scientific Electronic Library Online (SCIELO), Global Index Medicus (GIM) and PsycINFO) were systematically searched for peer-reviewed primary studies published from January 1980 to July 2025 in any language. A consistent search strategy to identify the risk and effect size of interest was used in all databases and across all risk–outcome associations (Supplementary Table 1). Studies were included if they reported biomarker-defined exposures (for example, anaemia defined by haemoglobin or haematocrit concentrations) and specified maternal and neonatal outcome(s) and

Table 3 | Reference definition of outcomes

Outcome	Reference definition
Maternal outcomes	
All-cause maternal mortality	Death due to obstetric complications or preexisting or coexisting disorders exacerbated by pregnancy up to 1 year after the end of pregnancy
Antepartum haemorrhage	Vaginal bleeding from any cause at or beyond 20 weeks of gestation and before the onset of labour
Postpartum haemorrhage	Vaginal bleeding of >500 ml following vaginal delivery or of >1,000 ml after caesarean delivery, regardless of the cause, within 24 h of delivery
Gestational hypertension	New onset of hypertension in a pregnant person after 20 weeks of gestation as defined by having a blood pressure of >140/90 mm Hg on ≥1 occasion
Pre-eclampsia	Hypertension (>140/90 mm Hg) on >1 occasion with proteinuria (excess protein in the urine (that is, ≥0.3 g protein per litre)) or signs of end-organ damage (including injury to the kidneys (for example, proteinuria, liver, platelets), lungs and brain (for example, headaches, changes in vision, seizures))
HDOP	Gestational hypertension and pre-eclampsia (as defined above) and HDOP
Maternal sepsis and other infections	•Sepsis during pregnancy, during labour and delivery, after an abortion or postpartum (up to 6 weeks after delivery), defined as an identified source of infection, clinical signs of infection (vital sign abnormalities) and evidence of organ dysfunction •Infections (without sepsis) are believed to have a close epidemiological relationship with pregnancy, including genitourinary tract infections (for example, urinary tract infection), obstetrical wound infections (for example, surgical site infection from caesarean section) and indications of infection such as postpartum fever.
Peripartum depression	Major depressive episode (involving the experience of either depressed mood or loss of interest/pleasure, for most of the day, nearly every day, for at least 2 weeks) occurring during pregnancy or up to a year following delivery (Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition criteria)
Neonatal outcomes	
All-cause neonatal mortality	Neonatal or perinatal death within 27 days of life, whereas early all-cause neonatal mortality is death within the first 7 days of life
All-cause stillbirth	Fetal death at or after 20 weeks of gestation
PTB	Live delivery at completed gestational age of <37 weeks
Moderate PTB	Live delivery at completed gestational age of between 32 and 37 weeks
Very PTB	Live delivery at completed gestational age of between 28 and 32 weeks
Extreme PTB	Live delivery at completed gestational age of <28 weeks
LBW	Live delivery with birth weight of <2,500 g
Very LBW	Live delivery with birth weight of <1,500 g
Extreme LBW	Live delivery with birth weight of <1,000 g
SGA	Live delivery with birth weight of <10th percentile for gestational age regardless of the reference population
LGA	Live delivery with birth weight of >90th percentile for gestational age regardless of the reference population
Neonatal sepsis	Infections during the neonatal period that advance to a systemic bloodstream infection (sepsis) or sepsis

Note: For stillbirth, the primary analysis used a threshold of ≥20 weeks gestation (Centers for Disease Control and Prevention/American College of Obstetricians and Gynecologists), with sensitivity analyses applying alternative thresholds (>22, >24 and >28 weeks) to align with other regional and international definitions, including the International Classification of Diseases, Royal College of Obstetricians and Gynaecologists and WHO standards.

measurable quantitative associations (that is, RR, odds ratio (OR) or hazard ratio). To ascertain temporality, we required exposure measurement before outcome onset. In addition to anaemia and low and high haemoglobin, we accepted iron deficiency and iron-deficiency anaemia as exposures if quantitatively defined by a biomarker threshold (for example, serum ferritin). Primary and secondary analyses of all study designs were eligible, excluding conference abstracts, case studies and modelling studies. Clinical trial studies were included if they reported associations adjusted for or stratified by trial arms. To ensure generalizability, we excluded populations with atypical anaemia prevalence or elevated outcome risk (for example, mothers with multiple pregnancies, HIV, malaria or chronic kidney diseases). We also excluded studies where mothers received anaemia treatments beyond standard antenatal care (for example, iron therapy, blood transfusion) to reflect haemoglobin levels before additional interventions. Our general and outcome-specific inclusion and exclusion criteria are outlined in Supplementary Tables 2 and 3. We used DistillerSR, a software that leverages natural language processing (NLP), to improve efficiency and facilitate priority screening of studies with higher inclusion likelihood[52,53]. Supplementary Information section 1.3 and Supplementary Fig. 1 detail the protocols of our NLP-facilitated title and abstract, as well as full-text, screening.

Information on publication, study design, study population, exposure/outcome, measures of association and confounder adjustment (Supplementary Tables 4 and 5) was extracted into a standardized template in Microsoft Excel or REDCap data collection tools[54,55]. Supplementary Information section 1.5 outlines the data extraction protocol. We manually calculated RRs (for a cohort analytical design) or ORs (for a case-analytical design) and uncertainties when studies reported sufficient raw data but no effect size. To ensure data quality, at least 10% of studies were screened and extracted in duplicate, with discrepancies resolved by consensus. Outcome-specific PRISMA diagrams are available in Supplementary Figs. 2–6.

The study followed a prespecified protocol, with contributing reviews registered in PROSPERO (CRD42022368239, CRD42023389050, CRD42022364816), covering study identification, eligibility criteria and primary outcomes. Analytical methods followed the standard Burden of Proof framework and are described below.

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Estimating the shape of the exposure–outcome association

We modelled the effect size (for example, RR or OR) across a range of exposures using the Burden of Proof methodology. For studies reporting multiple effect sizes for the same exposure in the same population, we prioritized the authors' primary adjusted model or otherwise the estimate with the most comprehensive confounder adjustment. This approach aims to reduce confounding without assuming that more covariates yield less biased estimates. Additionally, to account for autocorrelation from multiple observations within the same study and reduce the influence of non-mutually exclusive exposure groups, we adjusted the standard errors of the effect sizes by a factor proportional to the number of repeated measurements. We also imputed missing values for key variables in the Burden of Proof approach, such as age and haemoglobin concentration bounds. When age start and end were not reported, they were inferred from the mean and standard deviation (if reported) or imputed as 10 and 54, representing the female reproductive age range, for lower and upper bounds, respectively. For missing lower or upper bounds of haemoglobin concentrations, we used age-, sex-, location- and year-specific haemoglobin distributions derived from various population-based health surveys modelled in the GBD study, including the Demographic and Health Survey, Multiple Indicator Cluster Survey series and national micronutrient surveys[15]. Based on the sample population in each study, we then imputed population-weighted minimum and maximum haemoglobin levels. We further standardized haemoglobin thresholds and bounds by adjusting all reported values for elevation using the updated WHO formula[16,34], applied via population-weighted elevation data. Studies using outdated elevation adjustments were corrected, and all data were aligned to GBD age and location classifications to ensure consistency across sources. Furthermore, to standardize exposure classification, studies with misaligned haemoglobin thresholds were reclassified by inverting exposure and reference groups using a cutoff of 130 g l^{-1}. Supplementary Information section 2.3 provides details on age and haemoglobin range imputation and other post-extraction data processing.

To characterize the shape of the continuous dose–response risk curves illustrating the relationship between low or high haemoglobin and each outcome across different haemoglobin ranges, we avoided the conventional log-linear assumption. Instead, we modelled the RR (gold standard) or OR for each outcome as the dependent variable using Bayesian regularized spline meta-regression. Haemoglobin concentration (g l^{-1}) was the continuous independent variable, defined by a reference exposure range (for example, normal haemoglobin concentrations) and an alternative exposure range (for example, haemoglobin thresholds for mild, moderate and severe anaemia or high haemoglobin).

After conducting exploratory analyses for each outcome and testing various model settings, including the number and placement of knots along the exposure range and linearity constraints on the tails, we selected a final model to describe the nonlinear dose–response relationship. This model used a quadratic spline with three interior knots placed based on the density of data points, linearity on the left tail and no constraints on monotonicity. Placing knots based on the data distribution allowed the spline to be more flexible and responsive to variations in exposure ranges with dense data, resulting in stable risk curves, even for outcomes with fewer studies (that is, fewer than ten studies). For outcomes with a large number of studies, these density-based knot placements produced risk curves consistent with knots placed around anaemia thresholds of 70, 100 and 110 g l^{-1} (ref. 16). Linearity on the left, but not the right, tail helps avoid imposing excessive constraints on the influence of sparse high-haemoglobin data points. No monotonicity constraints were enforced, given evidence of non-monotonic, J-shaped relationships in which both low and high haemoglobin levels were associated with increased risk of adverse maternal and neonatal outcomes. The final ensemble model was robust to knot placement, combining 50 individual models with randomly placed knots, where weights were assigned based on measures of model fit (based on a likelihood metric) and total variation (captured by the highest derivative). To improve accuracy and minimize the influence of extreme data and potential publication bias, we used a likelihood-based approach to trim the 10% of estimates with the largest standardized residuals relative to their standard errors, ensuring that the model relied on the 90% most consistent data. For outcomes with >100 studies (that is, PTB and LBW), a lighter 5% trim was applied. With such large sample sizes, heterogeneity and outliers are less likely to dominate, so a smaller trim effectively dampens extreme influence while retaining the bulk of the evidence. Finally, uncertainties in the mean risk curves were estimated using a parametric bootstrap approach. Details on the fitting procedure, including trimming, ensemble generation and posterior uncertainty estimation, are documented in Zheng et al.[30].

An exception to the above-mentioned model setting was made for outcomes with fewer than three studies assessing the effects of high haemoglobin levels, where we applied linearity on both tails to stabilize the model fit at both extremes of the haemoglobin distribution. Furthermore, in these cases, if one or both high-haemoglobin studies were excluded under the 10% trimming rule, we disabled trimming altogether (that is, 0% trimming) to preserve the contribution of these limited but critical data points to the overall risk curve. Another exception was for cases with sparse data on the extreme low end of the haemoglobin distribution, where we applied a uniform prior with a mean of 0 and a very small standard deviation (s.d. = 0.0001) on the leftmost spline knot to reduce curve flexibility and prevent implausible risk reductions at very low haemoglobin levels due to data sparsity.

Testing and adjusting for systematic biases

We quantified sources of bias according to the GRADE criteria, which requires assessment of potential bias due to (1) the representativeness of the study population; (2) the method of exposure assessment; (3) the method of outcome determination; (4) risk of reverse causation; (5) control for confounding; and (6) the potential for selection bias[50]. Details of general and outcome-specific binary bias covariates are provided in Supplementary Tables 39 and 40. To address the variability in confounders adjusted for in the included studies, a composite confounder grouping was created. For example, adjustments for antenatal care, antenatal supplement intake, parity, gravidity, gestational age, newborn sex and birth spacing were collectively considered to control for confounding related to obstetric factors. Bias covariates with identical values across all data points were excluded, and, when multiple covariates had identical distributions, one was randomly selected. As described by Zheng et al.[30], the Burden of Proof method used a stepwise least absolute shrinkage and selection operator (Lasso) approach to systematically test the effect of bias covariates by shrinking less important ones towards zero and identifying those that were statistically significant (at a 0.05 threshold) in describing outcome variance for inclusion in the final model. This approach allows for bias adjustment in the final risk curve while minimizing the potential for overfitting.

Quantifying between-study heterogeneity and evaluating publication bias

To quantify residual between-study heterogeneity—differences across estimates from the studies included in the meta-analysis—a linear mixed-effects model was used to estimate study-level random slopes (gamma) and random intercepts to account for within-study correlation. The uncertainty of the heterogeneity was derived using Fisher's information matrix[56], which is sensitive to the number of studies, study design and reported uncertainty in the effect size estimates. To further identify publication and reporting bias, we visually inspected funnel plots for asymmetry and statistically tested for significant correlation between the standard error and the reported effect size using Egger's regression[57]. We noted potential publication bias in the data but did not correct for it.

Estimating the BPRF

Incorporating the selected bias covariates, a Bayesian meta-regression model predicted the mean risk function with 95% UIs. Specifically, for each increment of 0.1 g l^{-1} in haemoglobin concentration, 1,000 draws were generated with and without accounting for between-study heterogeneity, with the mean of these draws used to estimate the mean risk and the 2.5th and 97.5th draws defining the 95% UI at each haemoglobin level. Using the computed uncertainty estimates that accounted for between-study heterogeneity, we then derived the BPRF, defined as the 5th-quantile risk curve closest to an RR of 1 (the null), representing the conservative (smallest) adverse effect of suboptimal haemoglobin concentration on the corresponding outcome that is consistent with available evidence. We then generated an ROS from the log of the BPRF averaged over the data-dense exposure range, defined as the 15th to 85th percentiles of the haemoglobin concentration. Due to the distinct pathophysiology of low and high haemoglobin, we assessed ROS separately, analysing the 15th to 85th percentiles below and above the reference level—the haemoglobin concentration with the lowest outcome risk. A higher positive ROS indicates a larger average effect size across the continuum of risk and stronger evidence for the estimated relationship.

For ease of interpretation, the ROS can be further categorized into a star-rating system, ranging from one to five stars. A one-star rating (ROS ≤ 0.0) indicates risks for which the mean relationship is not statistically significant based on our conservative analysis, suggesting that there may be no true association between maternal haemoglobin level and the selected outcome. A five-star rating (ROS > 0.62) indicates that, using the most conservative interpretation of the evidence, the average exposure of haemoglobin concentration (from the 15th percentile to reference for low haemoglobin and from the reference to the 85th percentile for high haemoglobin) increases the risk of the selected outcome by greater than 85%. Positive ROS ranges between one and five stars were further divided as follows: 0.0–0.14, two-star rating; >0.14–0.41, three-star rating; and >0.41–0.62, four-star rating.

Model validation and sensitivity analyses

Using simulated experiments, the validity of the Burden of Proof methodology as a meta-analytic tool has been meticulously evaluated[30,48]. For the current study, as described above, we chose our model settings based on exploratory analyses testing variations in the number and placement of knots along the exposure range and linearity constraints on the tails. Final models were selected based on biological plausibility, expert understanding of the dose–response relationships, and optimal predictive performance and fit as indicated by low mean squared error and negative log-likelihood. Dose–response risk estimates were validated by plotting the mean risk function, along with its 95% UI, against the extracted dose-specific RR data. Additionally, sensitivity analyses were conducted to assess whether the observed relationships varied by key factors such as the trimester of haemoglobin measurement and analytical assumptions, including the use of alternative outcome definitions, a uniform reference haemoglobin concentration and adjustment for socioeconomic confounders. For stillbirth, the primary analysis used fetal death at ≥20 weeks of gestation (Centers for Disease Control and Prevention/American College of Obstetricians and Gynecologists), with sensitivity analyses applying alternative thresholds (>22, >24 and >28 weeks) to align with other regional and international definitions, including the International Classification of Diseases and WHO standards.

To further examine the plausibility and interpretability of model-derived reference haemoglobin concentrations, defined as the TMRELs identified by the model, we estimated an alternative 'functional' reference haemoglobin threshold using a curvature-based approach. This involved smoothing the normalized mean risk curve for each outcome and computing its first and second derivatives to capture the rate and direction of risk change. Functional reference haemoglobin levels were identified at points where the second derivative changed sign (indicating an inflection in the curve) and the first derivative approached zero (suggesting the risk was plateauing). If no such point was detected within a physiologically plausible, non-anaemic range (110–140 g l^{-1}), the model-derived TMREL was retained. Comparing functional and model-derived reference haemoglobin levels allowed us to evaluate whether statistical estimates aligned with biologically meaningful thresholds and to determine when reference values might be better interpreted as a range rather than a fixed point.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The findings from this study are supported by data from published literature available in public online repositories. Details on data sources can be found on the Burden of Proof visualization tool (https://vizhub.healthdata.org/burden-of-proof/). Study characteristics and citations for all input data used in the analyses are presented in Supplementary Tables 7–11 and Supplementary Information section 2.2.1, respectively. Data points in each analysis are available in Supplementary Tables 12–24.

Code availability

Code used for data processing and for running the Burden of Proof models is publicly available online at https://github.com/ihmeuw-msca/burden-of-proof/tree/main/risks/low_hemoglobin. Analyses were carried out using R version 4.0.5, Python version 3.10.9 and Stata version 17.

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Author contributions

Managing the estimation or publications process: N.G.A.N., T.S.N., A.A.H., S.I.H. and T.A.M. Writing the first draft of the manuscript: N.G.A.N., T.S.N., T.A.M. and N.J.K. Primary responsibility for applying analytical methods to produce estimates: N.G.A.N., D.K.R., C.J.T., H.J.T. and N.J.K. Primary responsibility for seeking, cataloguing, extracting or cleaning data and designing or coding figures and tables: N.G.A.N., D.K.R., H.A.T., I.O., H.J.T., B.L.S., H.T.C., T.S.N., A.M., J.M.M., E.C., K.F., N.P. and N.J.K. Providing data or critical feedback on data sources: N.G.A.N., D.K.R., H.J.T., M.A.D., Z.A.B., D.G., K.M.M., P.S.S., R.J.D.S., S.I.H., N.P. and N.J.K. Developing methods or computational machinery: N.G.A.N., D.K.R., C.J.T., K.H., A.M., J.M.M., R.J.D.S., A.A., P.Z., N.M.G., S.I.H., N.P. and N.J.K. Providing critical feedback on methods or results: N.G.A.N., D.K.R., C.J.T., K.H., Z.A.B., W.M.G., K.E., D.G., K.M.M., P.S.S., R.J.D.S., N.P. and N.J.K. Drafting the work or revising it critically for important intellectual content: N.G.A.N., D.K.R., H.A.T., I.O., H.J.T., T.S.N., E.C., M.A.D., Z.A.B., W.M.G., D.G., K.M.M., P.S.S., S.A.M., S.I.H., T.A.M., N.P. and N.J.K. Managing the overall research enterprise: N.G.A.N., A.A.H., K.F., S.I.H. and N.J.K.

Competing interests

D.G. has received research grants paid to his institution from CSL Vifor (for a study on Patient Blood Management in oncologic surgery patients, co-financed with the Italian Ministry of Health), outside the submitted work. He serves on the CSL Vifor Advisory Board (for hemochromatosis, a topic unrelated to this paper), for which he receives consulting fees. He has received honoraria from Pharmacosmos for a lecture on iron deficiency in inflammatory bowel diseases (unrelated to this paper) and travel support from Sanofi to attend EHA-2025. D.G. also serves as President-elect of BIOIRON (International Society for the Study of Iron in Biology and Medicine), an unpaid voluntary position. N.J.K. has received consulting fees from the Foundation for Health Care Quality, Fujifilm Sonosite and Philips Inc. unrelated to the present work. He has received travel expenses and an honorarium from Bristol Myers Squibb for presenting Global Burden of Disease results. K.M.M. has received grants from Agios Pharmaceuticals and Pharmacosmos. He serves on advisory boards and has received consulting fees from Novartis, Bristol Myers Squibb (Celgene Corp), Agios Pharmaceuticals, CRISPR Therapeutics, Vifor Pharma, Novo Nordisk and Pharmacosmos, all outside the submitted work. P.S.S. serves on the American Academy of Pediatrics Committee on Nutrition (unpaid). All other authors declare no competing interests.

Additional information

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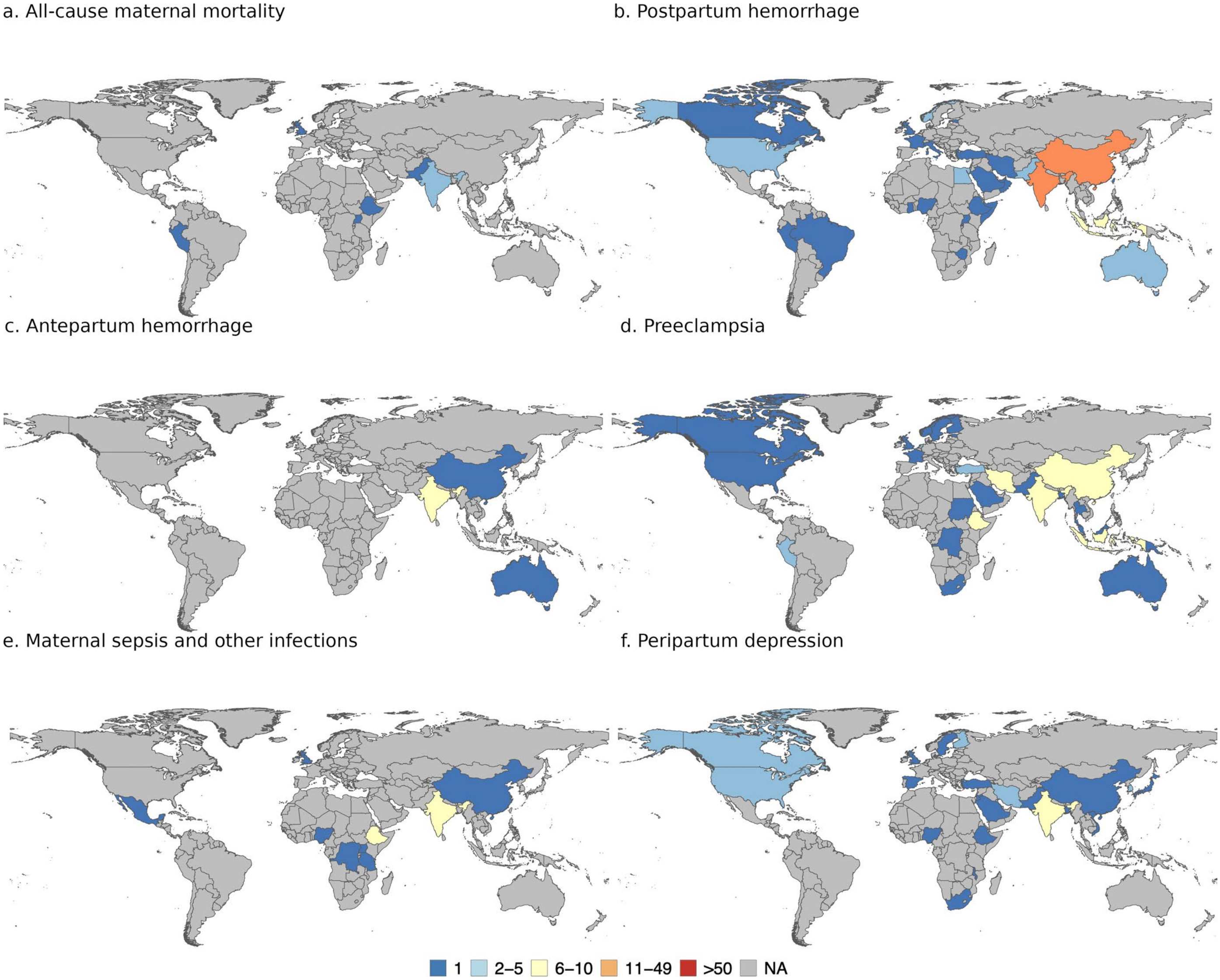

Extended Data Fig. 1 | Country-level counts of studies on maternal hemoglobin and maternal outcomes. Country-level counts of unique extracted studies on maternal hemoglobin and maternal outcomes: all-cause maternal mortality (**a**), postpartum hemorrhage (**b**), antepartum hemorrhage (**c**), preeclampsia (**d**), maternal sepsis and other infections (**e**), and peripartum depression (**f**). Map boundaries adapted from GADM.

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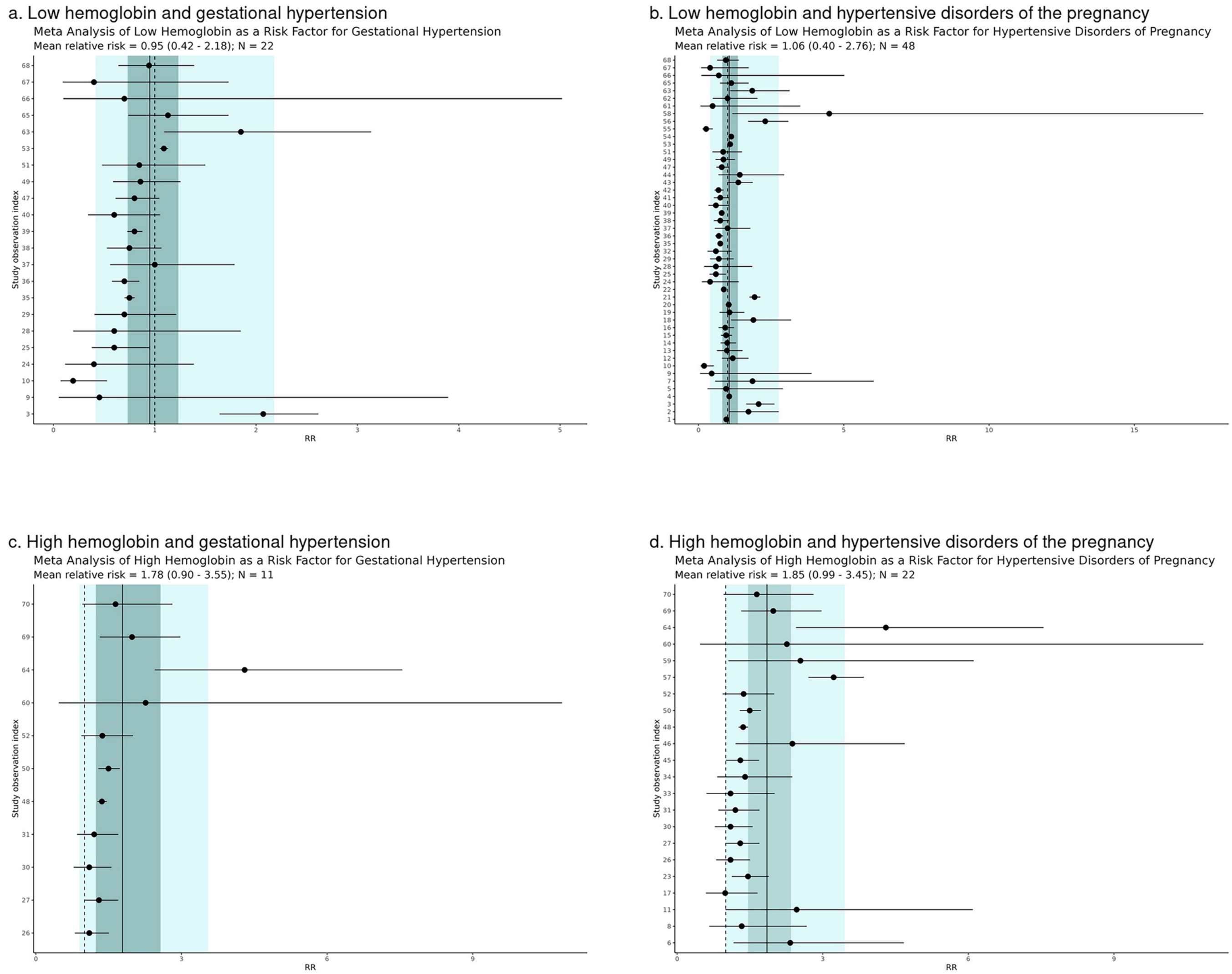

Extended Data Fig. 2 | Forest plot and pooled estimates of the association between binary maternal hemoglobin and pregnancy hypertension outcomes: low Hb and gestational hypertension (a), low Hb and hypertensive disorder of pregnancy (b), high Hb and gestational hypertension (c), and high Hb and hypertensive disorder of pregnancy (d). Note: N: the number of observations. Each black dot and horizontal line represents the reported relative risk (RR) and standard error from an individual observation. The vertical black line shows the mean estimated RR across studies. Light and dark shaded regions indicate 95% uncertainty intervals excluding and including between-study heterogeneity, respectively.

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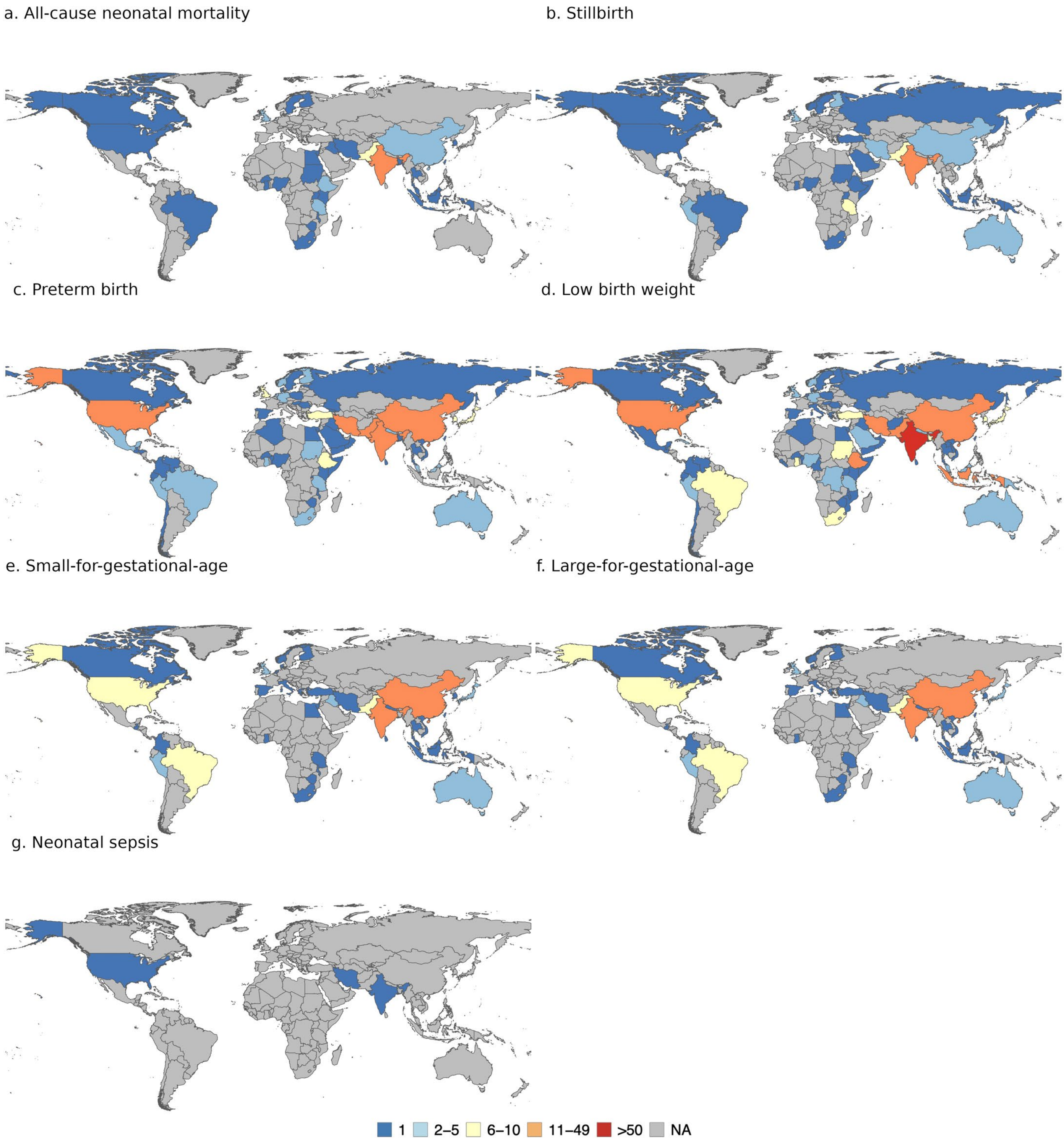

Extended Data Fig. 3 | Country-level counts of studies on maternal hemoglobin and neonatal outcomes. Country-level counts of unique extracted studies on maternal hemoglobin and neonatal outcomes: all-cause neonatal mortality (**a**), all-cause stillbirth (**b**), preterm birth (**c**), low birth weight (**d**), small for gestational age (**e**), large-for-gestational-age (**f**), and neonatal sepsis (**g**). Map boundaries adapted from GADM.

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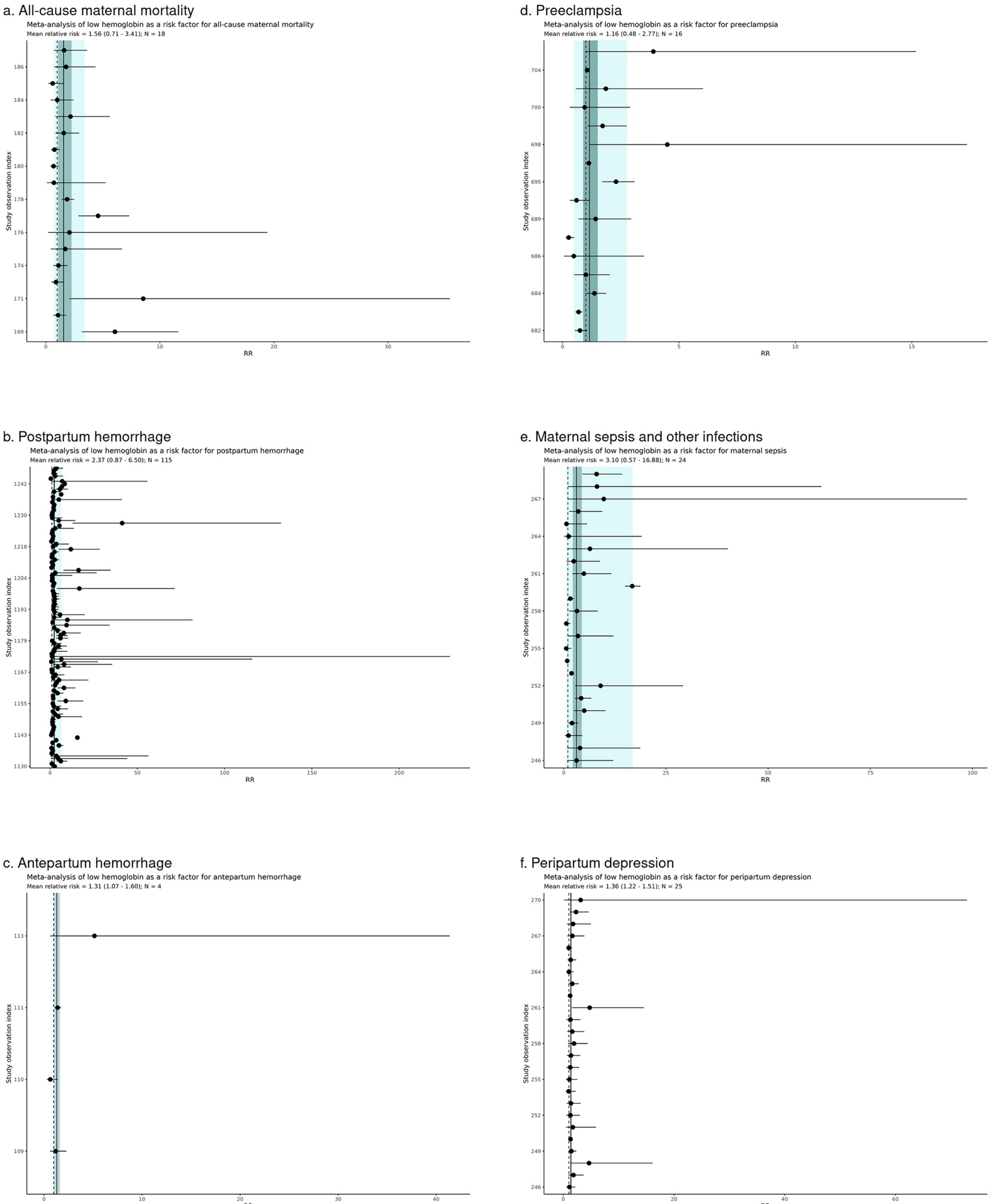

Extended Data Fig. 4 | Forest plot and pooled estimates of the association between binary maternal anemia and maternal outcomes: all-cause maternal mortality (a), postpartum hemorrhage (b), antepartum hemorrhage (c), preeclampsia (d), maternal sepsis and other infections (e), and peripartum depression (f). Note: N: the number of observations. Each black dot and horizontal line represents the reported relative risk (RR) and standard error from an individual observation. The vertical black line shows the mean estimated RR across studies. Light and dark shaded regions indicate 95% uncertainty intervals excluding and including between-study heterogeneity, respectively.

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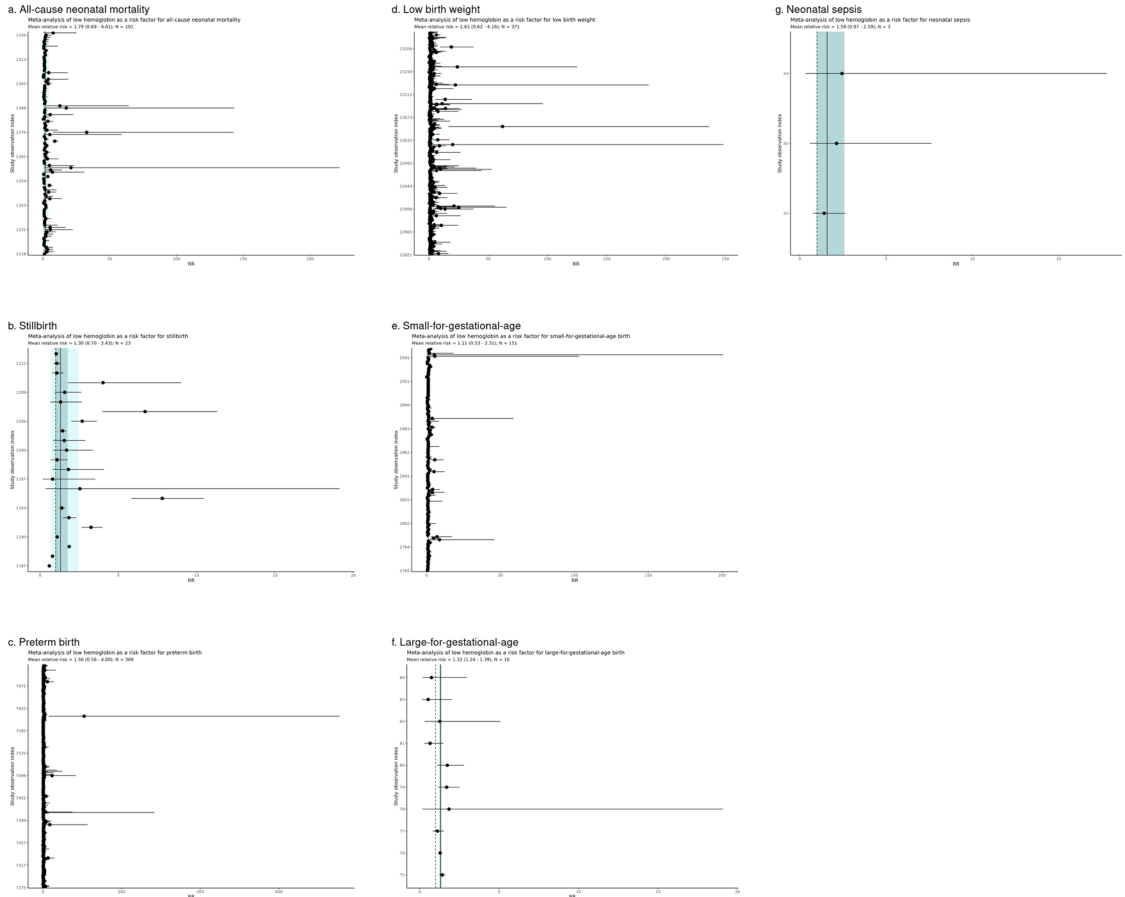

Extended Data Fig. 5 | Forest plot and pooled estimates of the association between binary maternal anemia and neonatal outcomes: all-cause neonatal mortality (a), all-cause stillbirth (b), preterm birth (c), low birth weight (d), small-for-gestational-age (e), large-for-gestational-age (f), and neonatal sepsis (g). Note: N: the number of observations. Each black dot and horizontal line represents the reported relative risk (RR) and standard error from an individual observation. The vertical black line shows the mean estimated RR across studies. Light and dark shaded regions indicate 95% uncertainty intervals excluding and including between-study heterogeneity, respectively.

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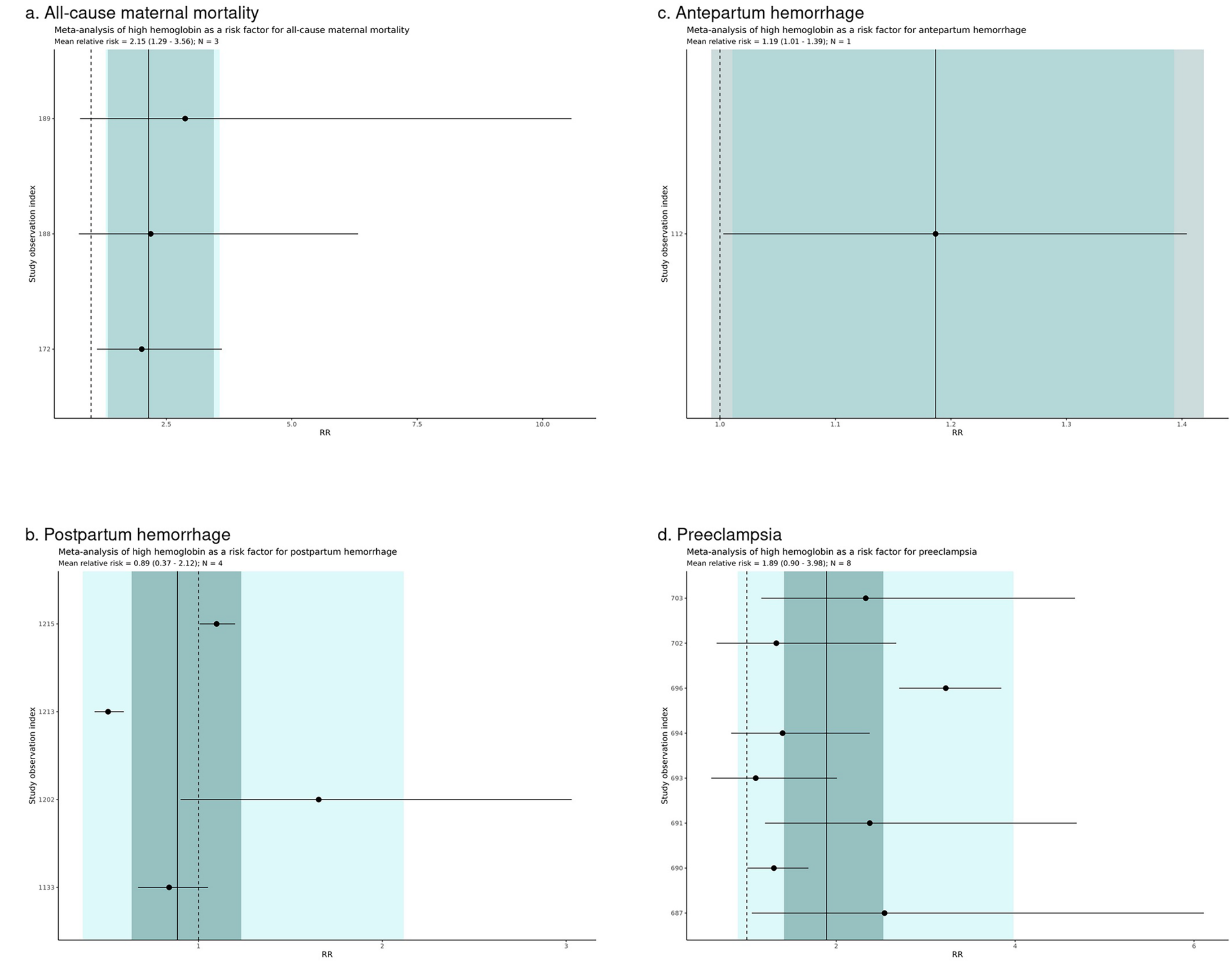
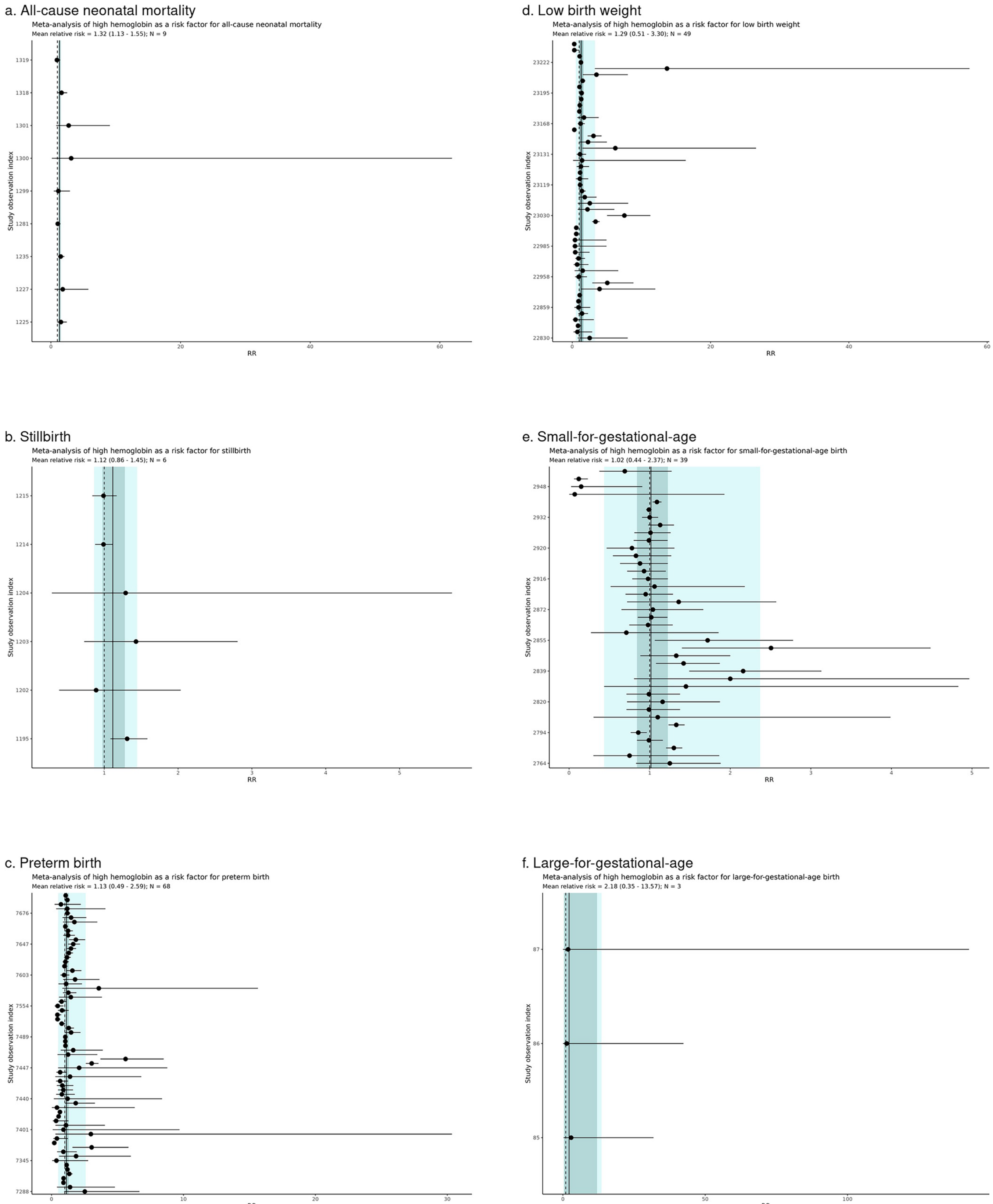


Extended Data Fig. 6 | Forest plot and pooled estimates of the association between binary maternal high hemoglobin and maternal outcomes: all-cause maternal mortality (a), postpartum hemorrhage (b), antepartum hemorrhage (c) and preeclampsia (d). Note: N: the number of observations. Each black dot and horizontal line represents the reported relative risk (RR) and standard error from an individual observation. The vertical black line shows the mean estimated RR across studies. Light and dark shaded regions indicate 95% uncertainty intervals excluding and including between-study heterogeneity, respectively.



Extended Data Fig. 7 | Forest plot and pooled estimates of the association between binary maternal high hemoglobin and neonatal outcomes: all-cause neonatal mortality (a), all-cause stillbirth (b), preterm birth (c), low birth weight (d), small-for-gestational-age (e), and large-for-gestational-age (f). Note: N: the number of observations. Each black dot and horizontal line represents the reported relative risk (RR) and standard error from an individual observation. The vertical black line shows the mean estimated RR across studies. Light and dark shaded regions indicate 95% uncertainty intervals excluding and including between-study heterogeneity, respectively.

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Extended Data Table 1 | Alternate and excluded definition of outcomes

Outcomes	Alternate case definitions	Exclusion
Maternal outcomes		
All-cause maternal mortality	• N/A	Accidental and incidental deaths
Antepartum hemorrhage	• Placental disease or disorders with blood loss • Placental abruption with blood loss • Placental previa or accreta with blood loss	N/A
Postpartum hemorrhage	• Postpartum hemorrhage with unspecified blood loss • Postpartum hemorrhage with unspecified timing of blood measurement (assumed to be within 24 hours)	• Blood transfusion without indication of postpartum hemorrhage • Postpartum hemorrhage occurring after 24 hours of birth
Gestational hypertension	New onset of hypertension in a pregnant person after 20 weeks gestation even when blood pressure measurement not specified	N/A
Preeclampsia	• Hypertension and proteinuria without any mention of signs of end-organ damage • Cases of preeclampsia without any mention of blood pressure, urine protein level or signs of organ damage	N/A
Hypertensive disorders of the pregnancy (HDOP)	N/A	N/A
Maternal sepsis and other infections	N/A	Sexually transmitted diseases
Peripartum depression	• Depression as defined by International Classification Disease (ICD) or Research Diagnostic Criteria (RDC) or previous versions of the DSM (e.g. DSM III, DSM IV, DSM IV-TR) • Major depressive episodes during pregnancy or up to 12 months after pregnancy (including after live births, stillbirths, miscarriages, or abortion) as assessed by diagnostic criteria and/or validated symptoms checklist such as Edinburgh Postnatal Depression Scale, Patient Health Questionnaire-9 and others	N/A
Neonatal outcomes		
All-cause neonatal mortality	• Neonatal or perinatal death within 28 days of life • Death within unspecified length of life described as neonatal or perinatal in nature	• Stillbirth • Death occurring after the 28^{th} day of life
All-cause stillbirth	• Fetal death at or after weeks of gestation beyond 20 weeks (e.g., 22, 24, 28 weeks) • A delivery described as stillbirth	N/A
Preterm birth (PTB)	• Live delivery at completed gestational age at 37 weeks • PTB status with unspecified gestational age cutoff	N/A
Moderate PTB	• Live delivery at completed gestational age between 32 and 37 weeks, inclusive of the weeks	
Very PTB	• Live delivery at completed gestational age between 28 and 32 weeks, inclusive of the weeks	
Extreme PTB	• Live delivery at completed gestational age <28 weeks, inclusive of the week	
Low birthweight (LBW)	• Live delivery with birthweight ≤2500g • LBW status with unspecified weight cutoff	N/A
Very LBW	• Live delivery with birthweight ≤1500 g	
Extreme LBW	• Live delivery with birthweight ≤1000 g	
Small-for-gestational-age (SGA)	• Live delivery with birthweight ≤10^{th} percentile for gestational age • SGA status with unspecified percentile cutoff	SGA defined using thresholds other than percentiles, such as Z-scores or standard deviations
Large-for-gestational-age (LGA)	• Live delivery with birthweight ≥90^{th} percentile for gestational age • LGA status with unspecified percentile cutoff	LGA defined using thresholds other than percentiles, such as Z-scores or standard deviations
Neonatal sepsis	N/A	N/A

Note: For stillbirth, the primary analysis used a ≥ 20-week gestational threshold (Centers for Disease Control and Prevention / American College of Obstetricians and Gynecologists), with sensitivity analyses applying alternative thresholds (>22, >24, and >28 weeks) to align with other regional and international definitions, including the International Classification of Diseases, Royal College of Obstetricians and Gynaecologists, and World Health Organization standards.

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Reporting Summary

Nature Portfolio wishes to improve the reproducibility of the work that we publish. This form provides structure for consistency and transparency in reporting. For further information on Nature Portfolio policies, see our Editorial Policies and the Editorial Policy Checklist.

Statistics

For all statistical analyses, confirm that the following items are present in the figure legend, table legend, main text, or Methods section.

n/a	Confirmed	
☐	☒	The exact sample size (n) for each experimental group/condition, given as a discrete number and unit of measurement
☐	☒	A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly
☐	☒	The statistical test(s) used AND whether they are one- or two-sided *Only common tests should be described solely by name; describe more complex techniques in the Methods section.*
☐	☒	A description of all covariates tested
☐	☒	A description of any assumptions or corrections, such as tests of normality and adjustment for multiple comparisons
☐	☒	A full description of the statistical parameters including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals)
☐	☒	For null hypothesis testing, the test statistic (e.g. F, t, r) with confidence intervals, effect sizes, degrees of freedom and P value noted *Give P values as exact values whenever suitable.*
☐	☒	For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings
☐	☒	For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes
☐	☒	Estimates of effect sizes (e.g. Cohen's d, Pearson's r), indicating how they were calculated

Our web collection on statistics for biologists contains articles on many of the points above.

Software and code

Policy information about availability of computer code

Data collection	To improve the efficiency of the screening of titles and abstracts, we utilized DistillerSR. Information on the publication (e.g., title, author, year of publication), study details (e.g., geographic location, years of data collection, study design), study population (e.g., age range, sex, geographic/population representativeness, sample size), exposure and outcome (e.g., definition, measurement and reporting methods, timing of measurement, severity [if applicable]), measures of association (e.g., RR and 95% confidence interval, count of outcomes in exposed and unexposed) and confounders adjustment were extracted into a standardized template in Microsoft Excel or REDCap data collection tools
Data analysis	Code used for data processing and for running the Burden of Proof models is publicly available online at https://github.com/ihmeuw-msca/burden-of-proof/tree/main/risks/low_hemoglobin. Analyses were carried out using R version 4.0.5, Python version 3.10.9, and Stata version 17.

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors and reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Portfolio guidelines for submitting code & software for further information.

Data

Policy information about availability of data

All manuscripts must include a data availability statement. This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A description of any restrictions on data availability
- For clinical datasets or third party data, please ensure that the statement adheres to our policy

The findings from this study are supported by data from published literature available in public online repositories. Details on data sources can be found on the Burden of Proof visualization tool (https://vizhub.healthdata.org/burden-of-proof/). Study characteristics and citations for all input data used in the analyses are presented in the Supplementary Tables 7-11 and Supplementary Information Section 2.2.1, respectively. Data points in each analysis are available in Supplementary Tables 12-24.

Research involving human participants, their data, or biological material

Policy information about studies with human participants or human data. See also policy information about sex, gender (identity/presentation), and sexual orientation and race, ethnicity and racism.

Reporting on sex and gender	This study did not involve direct recruitment of human participants. The analysis examined associations between maternal hemoglobin levels during pregnancy and stillbirth; therefore, the population of interest consisted of pregnant individuals as defined in the original studies. Sex- or gender-specific data were not analyzed beyond what was reported in the source studies.
Reporting on race, ethnicity, or other socially relevant groupings	Race, ethnicity, and other socially constructed groupings were not consistently reported across the primary studies included in the systematic review and were not analyzed in this study. Definitions and classifications of such variables, where present, were determined by the original study authors.
Population characteristics	This study synthesized data from previously published epidemiologic studies examining the association between maternal hemoglobin concentrations during pregnancy and stillbirth. Study populations consisted of pregnant individuals across multiple geographic regions and study designs, as described in the original publications.
Recruitment	No participants were recruited for this study. Data were extracted from previously published studies identified through a systematic literature review.
Ethics oversight	This study used aggregated data from previously published studies and did not involve identifiable individual-level data. Ethical approval was therefore not required. Ethical oversight for participant recruitment and data collection was obtained by the investigators of the original studies.

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Field-specific reporting

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☒ Life sciences ☐ Behavioural & social sciences ☐ Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see nature.com/documents/nr-reporting-summary-flat.pdf

Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size	No formal sample size calculation was performed because this study is a systematic review and meta-analysis of previously published observational studies. The sample size was determined by the number of eligible studies identified through systematic literature searches across eight databases (PubMed, Embase, Web of Science, CINAHL, CNKI, SCIELO, Global Index Medicus, and PsycINFO) from January 1980 to July 2025. All studies meeting the predefined inclusion criteria and reporting quantitative associations between maternal hemoglobin levels and selected maternal or neonatal outcomes were considered eligible. The final sample size reflects the number of studies and observations available for each outcome-specific dose–response meta-analysis.
Data exclusions	Inclusion and exclusion criteria were predefined in the study protocol and registered in PROSPERO (CRD42022368239, CRD42023389050, CRD42022364816). Studies were excluded if they lacked biomarker-defined exposure measurements, did not report relevant maternal or neonatal outcomes, or did not provide sufficient information to estimate an effect size. Additional exclusions included conference abstracts, case studies, modeling studies, and studies conducted in populations with atypical anemia prevalence or outcome risk (e.g., multiple pregnancies or populations with HIV, malaria, or chronic kidney disease). During modeling, predefined procedures such as trimming extreme residuals were applied to reduce the influence of outliers.
Replication	The study synthesizes findings from multiple independent observational studies conducted across diverse populations and settings. Replication is inherent to the meta-analytic design because effect estimates are derived from independent datasets. Analytical procedures followed the Burden of Proof framework using standardized statistical code and prespecified modeling assumptions. Risk estimates were

evaluated through model validation and sensitivity analyses examining alternative model specifications, exposure definitions, and outcome definitions.

Randomization	Randomization was not applicable because this study is a systematic review and meta-analysis of observational studies. Participants in the original studies were not randomized to exposure groups. Confounding was addressed in the included studies through statistical adjustment, and differences in adjustment strategies were evaluated through bias covariates incorporated into the meta-regression model.
Blinding	Blinding was not applicable to this study design. Data were extracted from previously published studies using predefined extraction protocols, and statistical analyses were conducted using aggregated study-level effect estimates.

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☒	☐ Antibodies
☒	☐ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☒	☐ Animals and other organisms
☐	☒ Clinical data
☒	☐ Dual use research of concern
☒	☐ Plants

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging

Clinical data

Policy information about clinical studies

All manuscripts should comply with the ICMJE guidelines for publication of clinical research and a completed CONSORT checklist must be included with all submissions.

Clinical trial registration	Not applicable. This study is a systematic review and meta-analysis of previously published studies and does not involve a prospective clinical trial or intervention requiring trial registration.
Study protocol	The study followed a prespecified protocol for systematic review and meta-analysis. Contributing reviews were registered in the International Prospective Register of Systematic Reviews (PROSPERO) under registration numbers CRD42022368239, CRD42023389050, and CRD42022364816. The protocol defined the search strategy, eligibility criteria, outcomes of interest, and analytical approach following the Burden of Proof framework.
Data collection	Data were obtained from previously published peer-reviewed studies identified through systematic searches of eight bibliographic databases (PubMed, Embase, Web of Science, CINAHL, CNKI, SciELO, Global Index Medicus, and PsycINFO) covering publications from January 1980 to July 2025. Eligible studies reported biomarker-defined maternal hemoglobin levels during pregnancy and their associations with maternal or neonatal outcomes. Data extraction included study characteristics, population information, exposure definitions, outcome definitions, and quantitative measures of association (e.g., relative risks, odds ratios, or hazard ratios). Because this study synthesized existing published data, no new participants were recruited and no primary clinical data were collected.
Outcomes	The study evaluated associations between maternal hemoglobin concentration during pregnancy and maternal and neonatal health outcomes. Maternal outcomes included all-cause maternal mortality, postpartum hemorrhage, antepartum hemorrhage, hypertensive disorders of pregnancy, maternal sepsis and other infections, and peripartum depression. Neonatal outcomes included all-cause neonatal mortality, stillbirth, preterm birth, low birthweight, small-for-gestational-age, large-for-gestational-age, and neonatal sepsis. Outcome definitions were based on those reported in the original studies and harmonized where possible. Associations were evaluated using the Burden of Proof meta-analytic framework to estimate dose–response relationships and conservative risk estimates.